

Groups

Office Hour One, rewritten for independent study

A self-contained account of the first office hour of *Office Hours with a Geometric Group Theorist* (Clay and Margalit). It follows the same route through the same ideas, in the same order, but every sentence, picture and problem here is written from scratch, and nothing is assumed beyond school algebra. You should be able to read it without the book beside you.

Background you need

What you need. Functions and what it means for one to be a bijection; how to compose two functions; sets and set notation; division with remainder. Nothing else. Matrices appear in one short example that can be skipped, and the words *ring* and *field* are never needed.

Big picture

Where this is going. A group is usually introduced as a set with a multiplication obeying three rules, which makes it look like a piece of bookkeeping. The point of this chapter is that it is nothing of the kind: every group is the collection of symmetries of some object, and a good choice of object turns algebraic questions into questions about a picture. The chapter ends with the construction that makes this precise, the *Cayley graph*, and with the statement that a group is exactly the symmetry group of its own Cayley graph.

How to read this

The coloured boxes are not decoration and they are not optional reading. *Why?* boxes contain the argument behind a claim that was only asserted. *Watch out* boxes contain the mistake that most readers make at that exact point. *Try it* boxes are short checks to do on paper before going on; they take a minute each and they are where the understanding actually happens. *Big picture* boxes say what a piece of machinery is for. The pictures use one convention throughout: grey labels mark *fixed positions*, and a coloured disc at a position names *the thing now sitting there*. So a disc that has drifted away from its grey label has moved, and a picture pair before and after a move shows exactly what the move did. Every figure is placed at the moment it is needed, and every figure is referred to in the text.

Contents

1	Symmetries of a cube	4
1.1	What counts as a symmetry	4
1.2	Counting the symmetries	4
1.3	The three families of axes	4
1.3.1	Axes through the centres of two opposite faces	5
1.3.2	Axes through two opposite corners	5
1.3.3	Axes through the midpoints of two opposite edges	6
1.3.4	The three families together	7
1.4	From turns to permutations	7

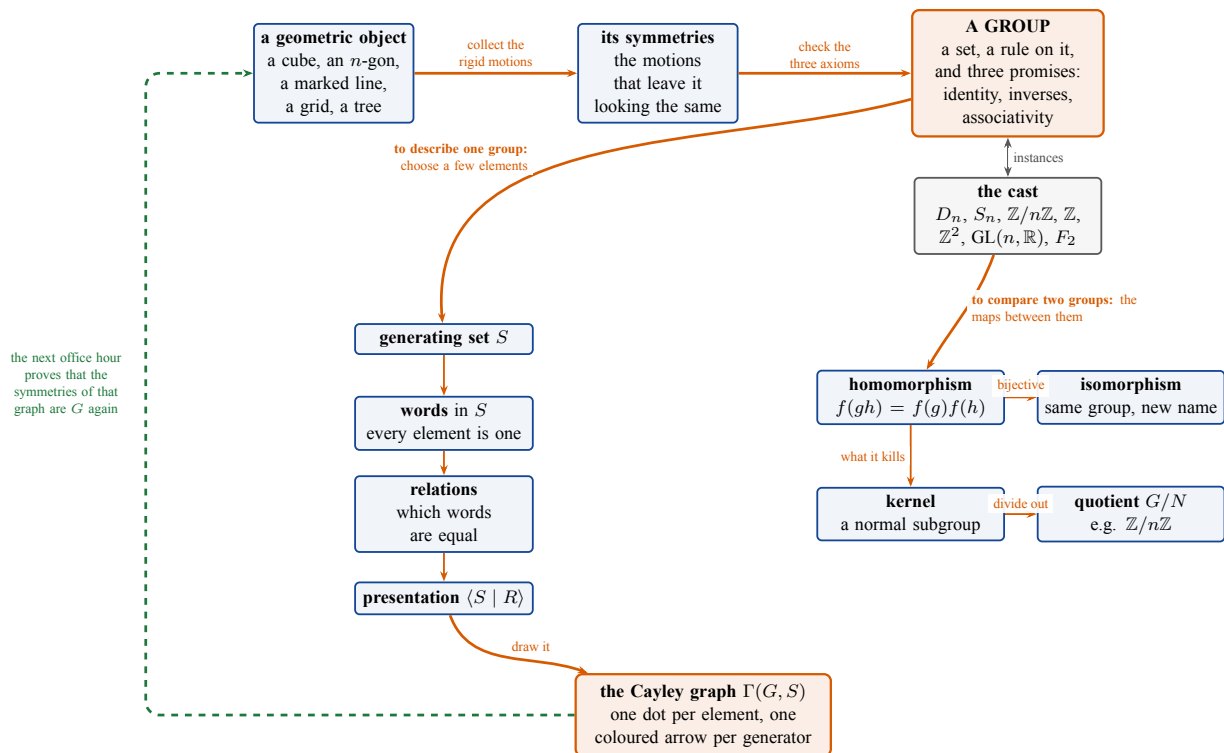


Figure 0.1 The whole chapter on one page. Read along the orange arrows: a geometric object gives up its symmetries, those symmetries turn out to obey three rules, and the rest of the chapter learns to describe such a thing on the left, to compare two of them on the right, and finally to draw one at the bottom. The dashed arrow is the loop the next office hour closes. Come back to this picture after each section.

2	What a group is	11
2.1	Multiplying symmetries	11
2.2	The definition	11
2.3	The three axioms on the cube	12
2.4	Why exactly these three axioms	14
2.5	Which of these are groups?	14
3	A gallery of groups	15
3.1	Dihedral groups: symmetries of a polygon	16
3.2	Symmetric groups: rearranging n things	18
3.3	The integers modulo n	23
3.4	Decorating a polygon to see $\mathbb{Z}/n\mathbb{Z}$	25
3.5	The integers as slides of a line	26
3.6	The plane lattice $\mathbb{Z}^2$	27
3.7	Matrix groups, in one paragraph	27
3.8	The free group of rank two	29

4	Generators and the Cayley graph	32
4.1	Generating sets	32
4.2	The Cayley graph	34
5	Comparing groups: homomorphisms	35
6	Subgroups, cosets and quotients	37
7	Presentations	38
8	Looking back: what each idea bought	40
8.1	What each idea buys	40
8.2	Algebra making geometry computable	41
9	Where this is going	42
10	Problems	43

1 Symmetries of a cube

Take a wooden cube, hold it in both hands, turn it, and put it back down in the space it started in. The cube looks exactly as it did. Whatever you just did to it is a *symmetry* of the cube. That is the whole idea, and the rest of this section is about taking it seriously enough to count the symmetries, to name them, and to multiply them together.

1.1 What counts as a symmetry

Two restrictions are hidden in the description above, and both matter.

First, the cube was moved *as a solid body*. Distances between its points never changed; it was not squashed, stretched or bent. A motion of space with that property is called *rigid*, and a symmetry of the cube is a rigid motion that carries the cube onto the same region of space it occupied before. Because it is rigid, such a motion sends corners to corners, edges to edges and faces to faces, and it preserves which corners are joined by an edge.

Second, the motion was carried out *with your hands*. You cannot turn a cube into its mirror image by turning it, however cleverly; you would have to pass it through a mirror. Motions you can physically perform are the ones that preserve the difference between left-handed and right-handed, and they are called *orientation-preserving*. Reflections are exactly the motions that fail this test.

Remark 1.1. Whether to allow reflections is a choice, not a fact, and different books make different choices. Allowing them doubles every count below: the cube has 24 rotations and 48 rigid symmetries altogether. Throughout this chapter, *symmetry of the cube* means rotation.

1.2 Counting the symmetries

Proposition 1.2. *A cube has exactly 24 rotational symmetries.*

Proof. Pick one corner of the cube and call it A . A symmetry must send A to a corner, so there are at most 8 choices for the image of A . Fix that choice. The three edges meeting at A must go to the three edges meeting at the image corner, in some cyclic order, and a symmetry is completely determined by where those three edges go. Spinning the cube about the long diagonal through the image corner cycles those three edges, and there are exactly 3 positions available. So each of the 8 choices admits exactly 3 symmetries, giving $8 \cdot 3 = 24$, and no symmetry is counted twice because the image of A together with the spin determines the motion. $\square$

Big picture

Knowing that the answer is 24 is much less useful than it looks. A group is not a number; it is a set with a multiplication. The interesting question is not how many symmetries there are but *how they combine*: which turn followed by which turn gives which. That question is what the rest of the chapter is built to answer, and the answer for the cube turns out to be surprisingly clean (Theorem 1.3).

1.3 The three families of axes

Every rotation other than “do nothing” turns the cube about a line through its centre, and a useful way to see all 24 is to sort those lines into three families. In each of the three figures below, one axis is drawn and

the turns about it are shown before and after; a second, smaller picture shows how many axes of that kind there are.

1.3.1 Axes through the centres of two opposite faces

Stand the cube on a table and spin it about the vertical line through the centres of the top and bottom faces. Quarter turns bring it back to the same region of space, so 90° , 180° and 270° are all symmetries; 360° is the identity and is counted separately. There are three such axes, one for each pair of opposite faces.

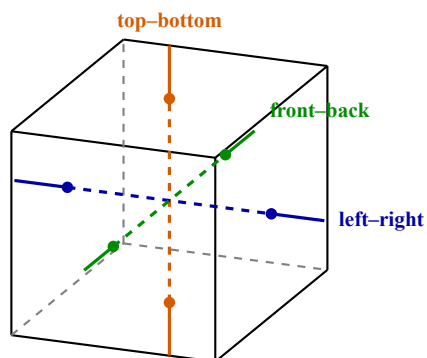


Figure 1.1 The three face axes, each joining the centres of two opposite faces, with the six face centres dotted and each axis drawn dashed where it passes through the inside of the cube. Three axes with three turns about each give $3 \cdot 3 = 9$ symmetries.

Reading Figure 1.2 row by row, with corners labelled as in Figure 1.3 below:

- *Quarter turn*, 90° . Each corner of the bottom face moves one place round, $A \rightarrow B \rightarrow F \rightarrow E \rightarrow A$, and the top face does the same, $D \rightarrow C \rightarrow G \rightarrow H \rightarrow D$. In the top view the four corners rotate one quarter of the way round the centre.
- *Half turn*, 180° . Every corner jumps to the opposite corner of its face: $A \leftrightarrow F$, $B \leftrightarrow E$, $C \leftrightarrow H$, $D \leftrightarrow G$. In the top view each corner swaps with the one diagonally across.
- *Three-quarter turn*, 270° , which is the same as -90° . The quarter turn performed the other way: $A \rightarrow E \rightarrow F \rightarrow B \rightarrow A$ and $D \rightarrow H \rightarrow G \rightarrow C \rightarrow D$. It is the inverse of the first row, which is why the two are listed as $\pm 90^\circ$ rather than as two unrelated moves.

1.3.2 Axes through two opposite corners

A **long diagonal** joins a corner to the opposite corner, passing through the centre of the cube. Balance the cube on one corner and spin: a third of a turn, 120° , brings it back, and so does 240° . There are four long diagonals, because the eight corners fall into four opposite pairs.

In Figure 1.4 the corners A and G never move, since they lie on the axis. The other six split into two rings of three. The 120° turn moves the lower ring one step, $B \rightarrow D \rightarrow E \rightarrow B$, and the upper ring one step, $C \rightarrow H \rightarrow F \rightarrow C$. The 240° turn is the same motion performed the other way, and the two are inverse to one another.

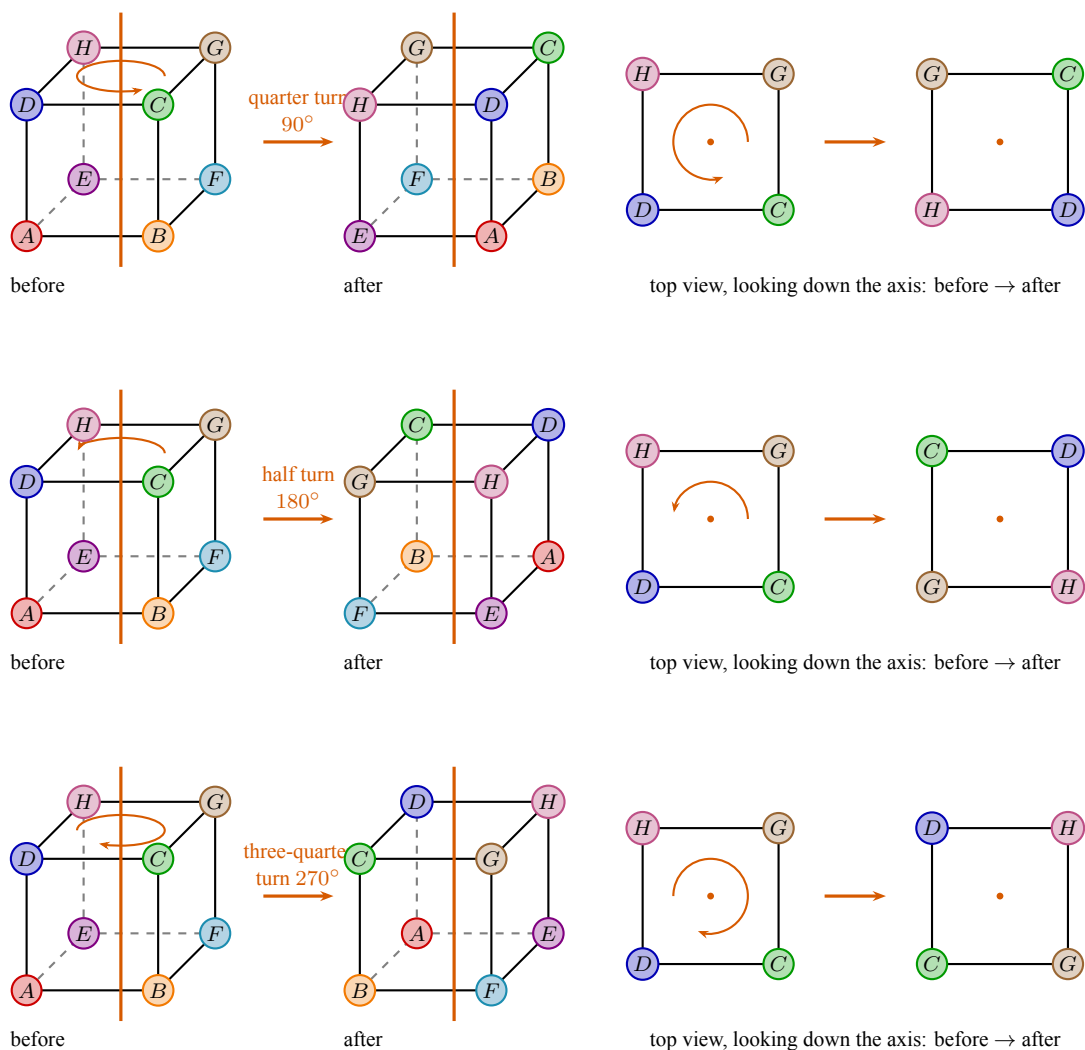


Figure 1.2 The three turns about one face axis, before and after, with the axis vertical. Beside each pair is the same turn seen from directly above, looking down the axis: the square of top corners D, C, G, H with the bottom corners hidden beneath them and the axis as the dot at the centre.

Why?

Why 60° fails. The top view is a hexagon, so a sixth of a turn looks plausible. But the six vertices of that hexagon alternate between two different heights along the axis: three of them are the corners next to G , and three are the corners next to A . A sixth of a turn would carry a corner next to G onto a corner next to A , that is, it would move a point to a point at a different distance along the axis, which no rotation about that axis can do. A third of a turn carries each ring to itself and is therefore available. In short, the corners of the cube supply only alternate vertices of the hexagon you think you see.

1.3.3 Axes through the midpoints of two opposite edges

Take an edge and the edge diametrically opposite it, and run a line through both midpoints. A half turn about that line is a symmetry, and nothing else about it is. The twelve edges make six opposite pairs, so there are six such axes.

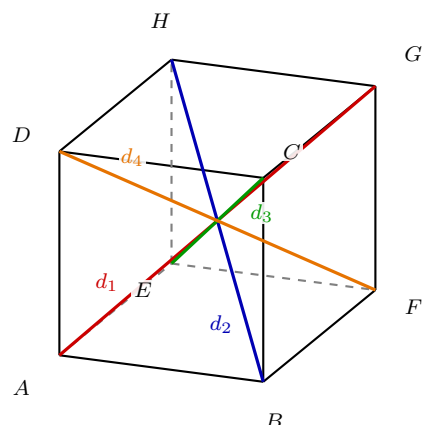


Figure 1.3 The four long diagonals, each joining two opposite corners through the centre. Four axes with two turns about each give $4 \cdot 2 = 8$ symmetries. The corner labels used throughout are A, B, C, D in front and E, F, G, H behind.

Two of these half turns deserve attention later: the axis meets the cube only at two edge midpoints, so no corner is fixed, and every corner moves. That is what makes this family the one that produces a single swap of diagonals in Section 2.

1.3.4 The three families together

kind of axis	how many axes	turns about each	symmetries in all
through two opposite face centres	3	$90^\circ, 180^\circ, 270^\circ$	9
through two opposite corners	4	$120^\circ, 240^\circ$	8
through two opposite edge midpoints	6	180°	6
and the identity	—	0°	1
total			24

The total agrees with Proposition 1.2, which is a genuine check: the two counts were made in completely different ways.

Try it

Hold a cube, or imagine one. Do the 120° turn about a long diagonal and then the 240° turn about the same diagonal, and confirm that the cube is back where it started. Then do a quarter turn about a vertical face axis followed by a quarter turn about a horizontal one, and do the same two turns in the other order. The results are different. Keep that fact; it is the first evidence that the multiplication we are about to define is not like multiplication of numbers.

1.4 From turns to permutations

Here is the observation that turns the cube from a bag of 24 motions into an algebraic object.

The cube has four long diagonals. Label them d_1, d_2, d_3, d_4 . A symmetry sends corners to corners and keeps opposite corners opposite, so it sends each long diagonal to a long diagonal: it produces a **rearrangement** of the four labels. Rearrangements of four things are much easier to handle than motions of space, and there are $4! = 24$ of them, which is suspicious.

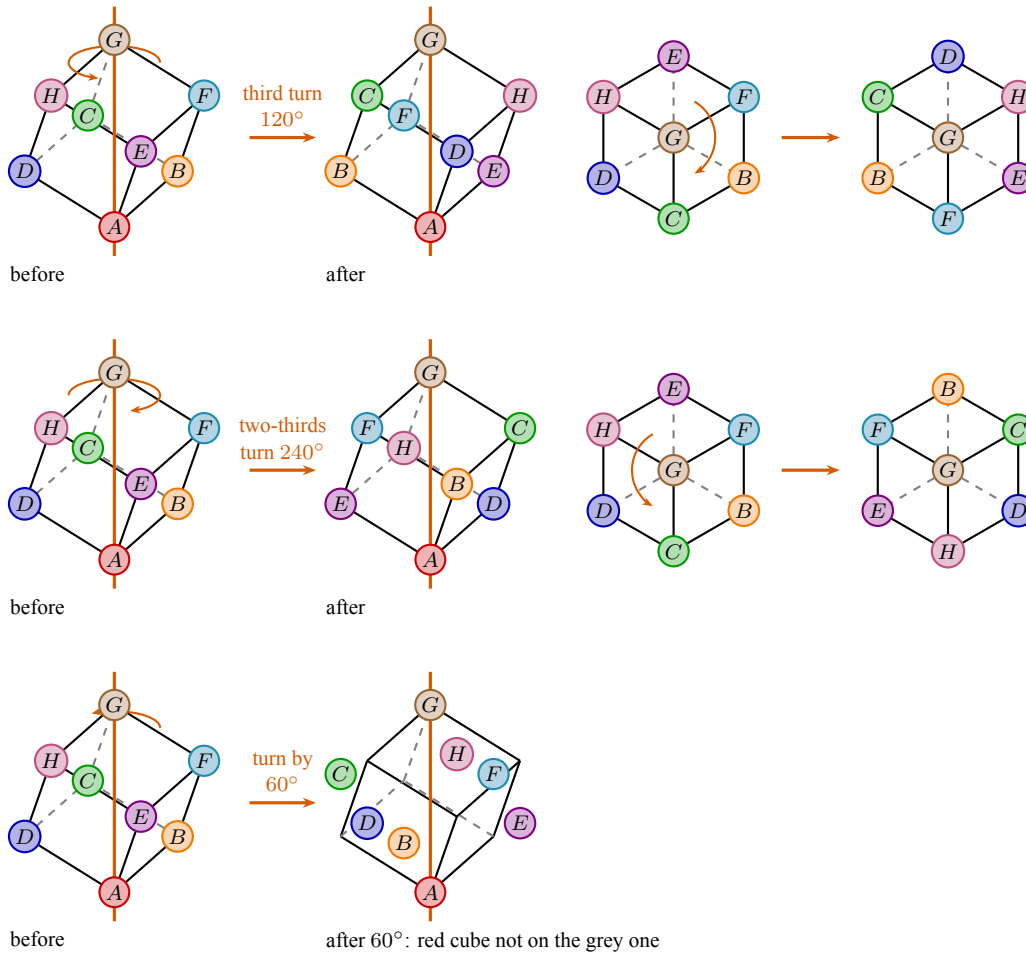


Figure 1.4 The turns about the long diagonal AG , before and after, with the cube standing on the corner A so that the axis is vertical. Beside each of the first two pairs is the view straight down the diagonal from above G : a regular hexagon whose six vertices are the six corners other than A and G , alternating between the three next to G and the three next to A . The third row attempts a turn by 60° and fails: the turned cube, in colour, does not sit on the original, in grey.

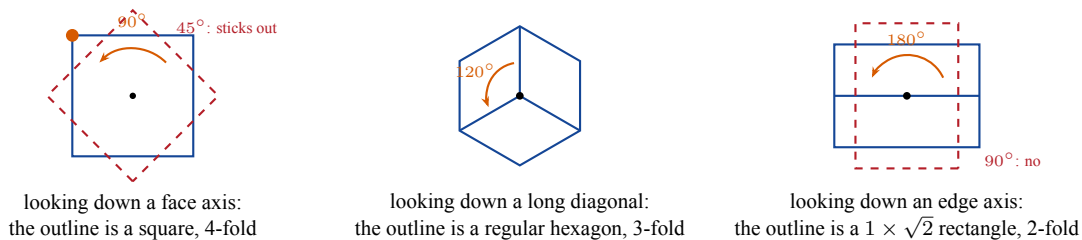


Figure 1.5 Why each kind of axis has the order it does. Looking straight down a face axis, the outline of the cube is a square, which comes back to itself after a quarter turn. Down a long diagonal the outline is a regular hexagon, but its six vertices alternate between two heights, so only a third of a turn is available. Down an edge axis the outline is a rectangle, and only a half turn works.

Theorem 1.3. *The map that sends a rotation of the cube to the rearrangement it induces on the four long diagonals is a bijection from the 24 rotations to the 24 rearrangements of four objects.*

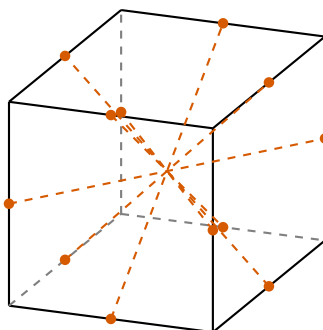


Figure 1.6 The six edge axes, dashed, with the twelve edge midpoints dotted. Each axis joins the midpoints of two opposite edges, and the twelve edges make six opposite pairs. Six axes with one turn about each give 6 symmetries.

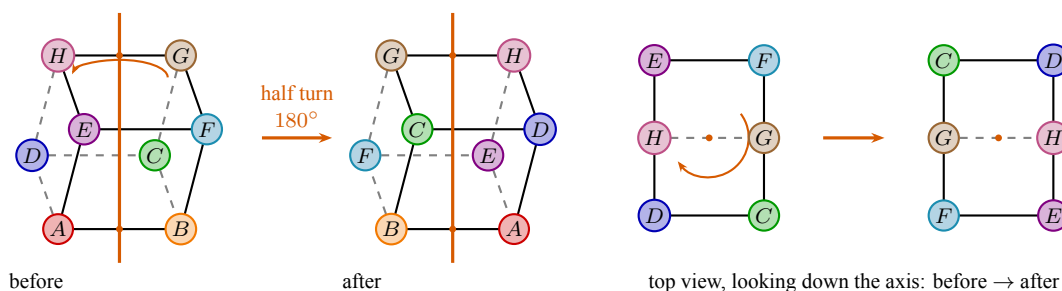


Figure 1.7 The half turn about the axis through the midpoints of AB and GH , before and after, with the cube standing on the edge AB . Every corner swaps with a partner: $A \leftrightarrow B$, $G \leftrightarrow H$, $C \leftrightarrow E$ and $D \leftrightarrow F$.

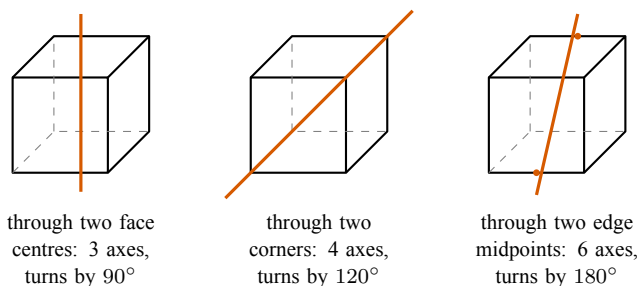


Figure 1.8 The three kinds of axis in one picture, for reference.

Proof. Both sets have 24 elements, by Proposition 1.2 and by $4! = 24$, so it is enough to show that every rearrangement occurs.

Consider first the rearrangements that swap two diagonals and leave the other two alone. Take $d_1 = AG$ and $d_2 = BH$. The edge AB joins an endpoint of each, and the opposite edge GH joins the other two endpoints. The half turn about the axis through the midpoints of AB and GH exchanges A with B and G with H , hence exchanges d_1 with d_2 , and one checks that it fixes d_3 and d_4 . The same argument works for any pair of diagonals, since any two of the four have a pair of opposite edges joining their endpoints in this way. So all six swaps occur.

Now, any rearrangement of four objects can be achieved by a succession of swaps: move the right object into the first place, then the right one into the second place, and so on. Performing the corresponding half turns one after another realises any rearrangement. Hence the map is onto, and being onto between two

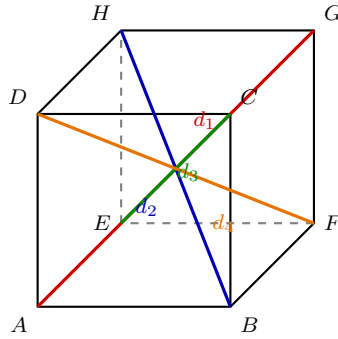


Figure 1.9 The cube with its four long diagonals in colour: $d_1 = AG$, $d_2 = BH$, $d_3 = CE$, $d_4 = DF$. Corners A, B, C, D are in front, E, F, G, H behind.

sets of the same finite size, it is a bijection. □

Remark 1.4. The proof used the fact that a succession of symmetries is again a symmetry, which is obvious for physical motions and will be given a name in the next section: *closure*. It also used that every rearrangement is a succession of swaps, which is proved carefully in Section 3.2.

Theorem 1.3 is worth more than the count it repeats. Rearrangements are functions, so they can be composed mechanically, on paper, with no geometric imagination at all. Once the dictionary is set up, a question about turning a cube becomes a question about composing two functions on a four-element set. The table below is that dictionary, family by family; notation for the right-hand column comes in Section 3.2, and it is worth returning to this table once that notation is available.

rotation of the cube	how many	effect on $d_1, \dots, d_4$	count
do nothing	1	leaves all four alone	1
quarter turn about a face axis, either direction	6	cycles all four	6
half turn about a face axis	3	two swaps at once	3
third of a turn about a long diagonal, either direction	8	fixes one, cycles three	8
half turn about an edge axis	6	swaps two, fixes two	6
total	24		24

Why?

Why a third of a turn about d_1 fixes d_1 and cycles the other three. The axis of the motion is the diagonal d_1 itself, so its two endpoints stay put and d_1 goes to d_1 . The three corners next to A are permuted cyclically by the turn, and the three diagonals starting at those corners are d_2, d_3, d_4 , so those three are cycled. This is also why the count in that row is 8 and not 4: each of the four diagonals gives two such turns, one in each direction.

Try it

Check the counts in the two right-hand columns of the table line by line. Then take the two half turns about perpendicular edge axes, perform one after the other, and work out which row of the table the result belongs to. (You will get a rotation of order three; find its axis.)

2 What a group is

2.1 Multiplying symmetries

Symmetries of the cube can be combined: do one, then do another, and the result is again a symmetry. That operation is the multiplication we are going to axiomatise, and it needs a convention, because “do g then h ” and “do h then g ” are different motions.

Definition 2.1. For symmetries g and h , the product gh means: *first do h , then do g .*

The order looks backwards until you notice where it comes from. A symmetry is a function on the points of space, and functions are written to the left of their inputs: $(g \circ h)(x) = g(h(x))$, so h acts first. Every computation in this chapter depends on this convention, so it is worth saying out loud once: *the right-hand factor acts first.*

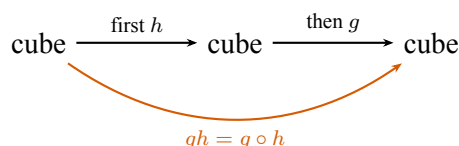


Figure 2.1 The convention in a picture: the product of two symmetries is the single symmetry that does the same job as the two in succession.

2.2 The definition

Definition 2.2. A **multiplication** on a set G is a function that takes an ordered pair of elements of G and returns an element of G , written $(g, h) \mapsto gh$. A **group** is a set G together with a multiplication on it such that:

- (G1) *Identity.* There is an element $1 \in G$ with $1g = g1 = g$ for every $g \in G$.
- (G2) *Inverses.* For every $g \in G$ there is an $h \in G$ with $gh = hg = 1$. It is written g^{-1} .
- (G3) *Associativity.* $(fg)h = f(gh)$ for all $f, g, h \in G$.

Two things about the definition are easy to read past.

The first is that the *multiplication is part of the data*. The same set can be a group with one rule and fail to be one with another: the integers with addition form a group, the integers with subtraction do not. So a group is a pair, a set and a rule, and naming only the set is an abbreviation.

The second is that the word “multiplication” is a generic label. For the integers it will mean addition; for symmetries it means composition; for words it will mean writing one after the other. Nothing about it needs to resemble multiplying numbers. In particular **nothing in Definition 2.2 says $gh = hg$** . Groups in which that happens to hold for every pair are called **abelian**, and most of the interesting groups in this chapter are not.

Remark 2.3. The requirement that gh lie back in G is often called **closure**. In Definition 2.2 it is not a separate axiom, because it is built into the word “function into G ”. When you are handed a candidate group, however, closure is usually the first thing to check and the easiest to get wrong; Section 2.5 has an example where it fails inside the cube.

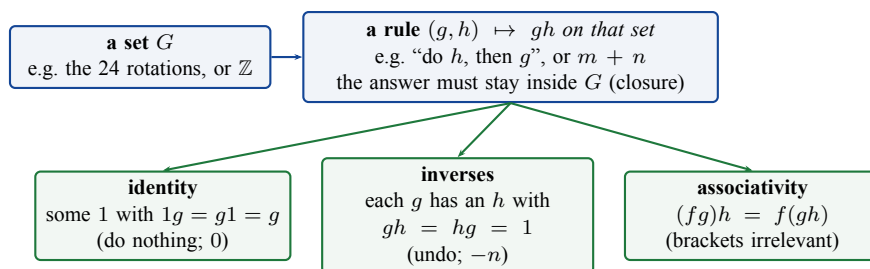


Figure 2.2 The anatomy of the definition: a set, a rule on that set whose answers stay inside the set, and three promises about the rule.

Remark 2.4. The symbol 1 in (G1) is a *name*, not the number one. For the integers under addition the identity is 0; for the cube it is “leave it alone”; for invertible matrices it is the identity matrix. Many books write e instead, from the German *Einselement*. Likewise g^{-1} is a name for the element that undoes g , and it is not “ $1/g$ ” unless the group happens to consist of numbers under ordinary multiplication.

2.3 The three axioms on the cube

The cube is the example to hold in mind, so here is each axiom checked on it. Grey letters are fixed positions, and a coloured disc names the corner of the original cube now sitting there.

(G1) Identity

“Leave the cube alone” is a rotation, by 0° . Doing it after any move r leaves the picture of r untouched, and doing it before r produces r as well, so $1r = r1 = r$.

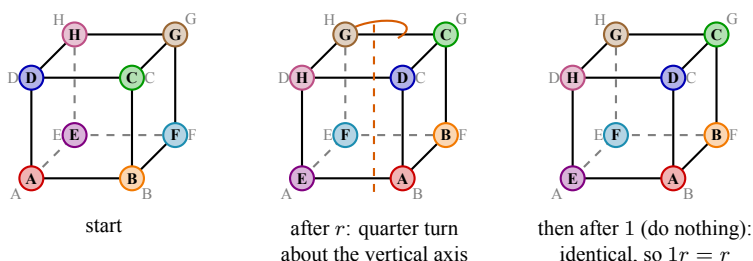


Figure 2.3 Identity on the cube: start, after a quarter turn r , then after doing nothing. The last two pictures are identical.

(G2) Inverses

The inverse of a quarter turn is the quarter turn the other way about the same axis. Three more turns in the original direction do the same job, so $r^{-1} = r^3$. A half turn is its own inverse, which is the analogue of 0 being its own inverse in the integers.

(G3) Associativity

Let h be a quarter turn about the vertical axis, g a quarter turn about the left–right axis, and f a half turn about the vertical axis. Performed one after another they do this:

Now $(fg)h$ says: do h , then the *single* rotation that f and g amount to together. And $f(gh)$ says: do the single rotation that g and h amount to, then f . Both routes end at the same cube as the row above.

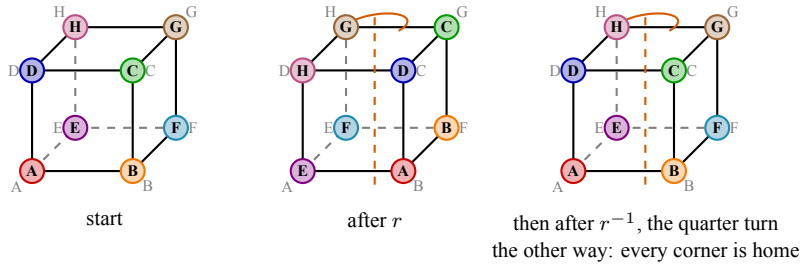


Figure 2.4 Inverses on the cube: start, after r , then after r^{-1} . Every disc is back on its grey letter, so $r^{-1}r = rr^{-1} = 1$.

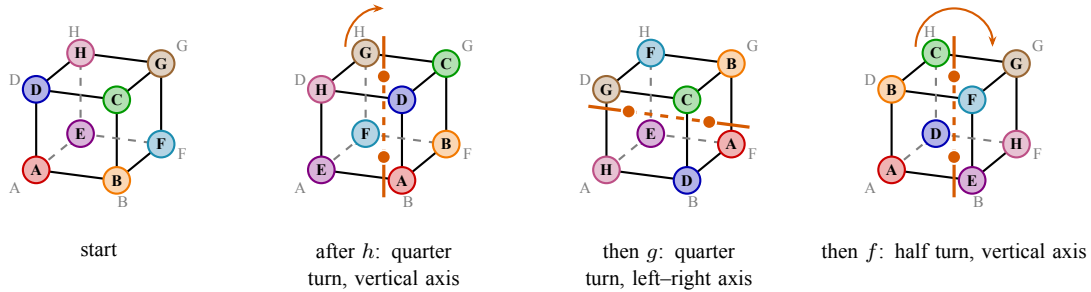


Figure 2.5 The three moves h , then g , then f , one picture at a time.

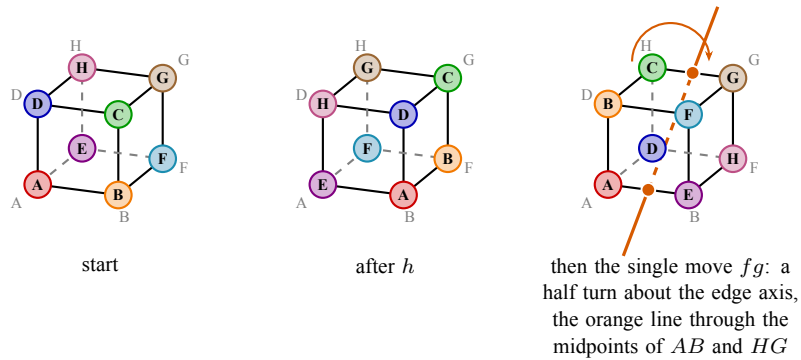


Figure 2.6 $(fg)h$: after h , the single move fg turns out to be the half turn about the edge axis through the midpoints of AB and HG .

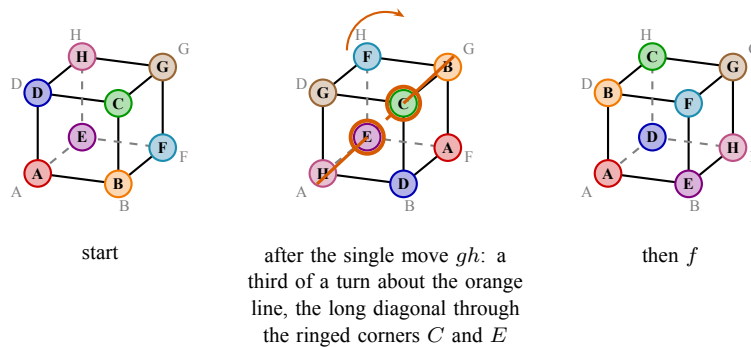


Figure 2.7 $f(gh)$: the single move gh is a third of a turn about the diagonal CE , and then f . Same final cube.

The brackets only decide which two moves you merge into one before performing anything; the cube still receives h , then g , then f , so the final position cannot depend on them. Said in one line: symmetries are functions, and composition of functions satisfies $((fg)h)(x) = f(g(h(x))) = (f(gh))(x)$ for every point x .

Big picture

This is worth remembering as a labour-saving device. *Any* group whose elements are functions and whose multiplication is composition satisfies (G3) automatically. Associativity only costs something when the multiplication is given by a formula with cases, as for the integers mod n in Section 3.3, or by a recipe on strings, as for the free group in Section 3.8.

2.4 Why exactly these three axioms

The three promises are not a random selection. They are exactly what is needed to solve equations. Suppose g and k are known and you want x with $gx = k$; on the cube, “which move x , followed by g , produces k ?”

$gx = k$	given	
$g^{-1}(gx) = g^{-1}k$	multiply both sides on the left by g^{-1}	(G2)
$(g^{-1}g)x = g^{-1}k$	regroup	(G3)
$1x = g^{-1}k$	$g^{-1}g = 1$	(G2)
$x = g^{-1}k$		(G1)

Each axiom is used exactly once, and the solution is unique, since the calculation derives it. Drop any one axiom and the argument breaks. The same three steps prove the **cancellation law**: if $gx = gy$ then $x = y$, in every group.

Try it

Solve $gx = k$ on the cube with g the quarter turn about the vertical axis and k the half turn about the same axis. Then solve $xg = k$ for the same g and k , and notice that the answer requires multiplying on the *right*, giving $x = kg^{-1}$. In an abelian group these two answers agree; in general they do not.

Proposition 2.5. *In any group the identity is unique, inverses are unique, and $(gh)^{-1} = h^{-1}g^{-1}$.*

Proof. If 1 and $1'$ both satisfy (G1) then $1 = 1 \cdot 1' = 1'$. If h and h' are both inverses of g then $h = h1 = h(gh') = (hg)h' = 1h' = h'$. For the last claim, multiply gh by $h^{-1}g^{-1}$ and regroup: $(gh)(h^{-1}g^{-1}) = g(hh^{-1})g^{-1} = gg^{-1} = 1$, and similarly on the other side. $\square$

The formula $(gh)^{-1} = h^{-1}g^{-1}$ is the **socks-and-shoes rule**: to undo “socks, then shoes” you take the shoes off first. It will be used constantly from Section 3.8 onwards.

2.5 Which of these are groups?

The fastest way to get used to Definition 2.2 is to watch it fail.

set	rule	closed	identity	inverses	associative
$\mathbb{Z}$	+	✓	0	$-n$	✓
$\{0, 1, 2, \dots\}$	+	✓	0	✗ no -5	✓
$\mathbb{Z}$	-	✓	✗ $0 - 5 \neq 5$	—	✗
$\mathbb{Z}$	$\times$	✓	1	✗ no $\frac{1}{2}$	✓
$\{1, -1\}$	$\times$	✓	1	each its own	✓
$\mathbb{R}$	$\times$	✓	1	✗ none for 0	✓
$\mathbb{R} \setminus \{0\}$	$\times$	✓	1	$1/x$	✓
the 24 rotations of a cube	do one, then the other	✓	do nothing	undo	✓
the 6 quarter turns of a cube	do one, then the other	✗	—	—	—
$\{1\} \cup \{3 \text{ face half turns}\}$	do one, then the other	✓	1	each its own	✓

Four of these ten are groups. The failures repay study. For $\mathbb{Z}$ with subtraction, $5 - 0 = 5$ but $0 - 5 \neq 5$, so 0 is an identity on one side only, and $(8 - 4) - 2 = 2$ while $8 - (4 - 2) = 6$, so (G3) fails too. For the non-negative integers, the only problem is inverses; everything else is fine, which shows the axioms are genuinely independent of one another. The last line but one is the interesting failure, and it happens inside the cube.

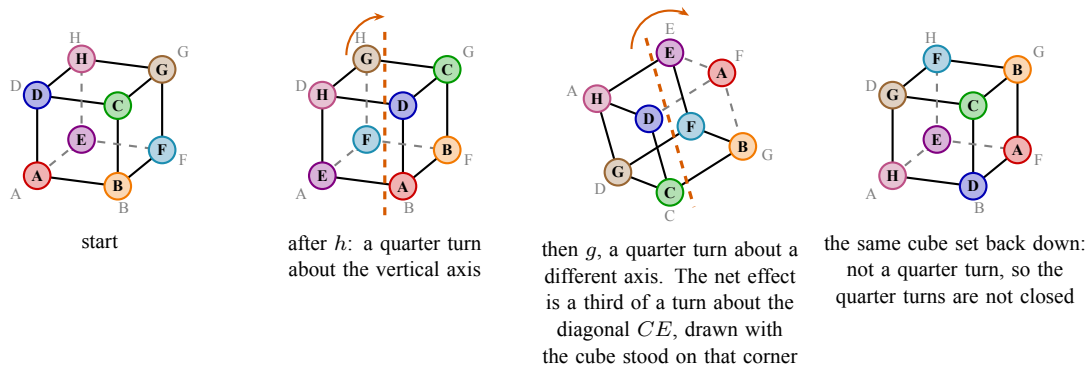


Figure 2.8 Closure can fail inside the cube: a quarter turn followed by a quarter turn about a different axis is a third of a turn about a long diagonal, not a quarter turn. So the six quarter turns are not closed under the multiplication, and they do not form a group on their own.

Why?

Associativity never fails for a set of symmetries, so it is not worth checking there; it fails only for rules invented by formula, like subtraction. This is why the checklist above has no ✗ in its last column except on the row for subtraction.

Try it

Decide, using the columns of the table: the even integers with +; the odd integers with +; the eight third-of-a-turn rotations of the cube together with the identity (try two of them about different diagonals); the four rotations about a single face axis.

3 A gallery of groups

Definition 2.2 is short, so the way to make it mean something is to run it against examples until the pattern is automatic. For each example below the routine is the same: say what the *set* is, say what the *rule* is, check closure and the three axioms, and only then collect the properties that make the example worth knowing.

3.1 Dihedral groups: symmetries of a polygon

The set. All rigid motions of a regular n -gon in the plane that carry it onto itself. There are $2n$ of them: n rotations, by multiples of $2\pi/n$ including the identity, and n reflections. The group is written D_n .

The rule. Do one, then the other, exactly as for the cube.

Reflections are allowed here, and it is worth being clear why this is not inconsistent with Section 2. A flat polygon lives in the plane but sits in space, so you can pick it up, turn it over and put it back; the motion is something your hands can do. For the solid cube there is no such trick.

Closure

Two reflections in a row are not a reflection, but they are certainly a symmetry, so nothing leaves the set. The same picture makes a second point that will matter later: the order of the two reflections changes the answer.

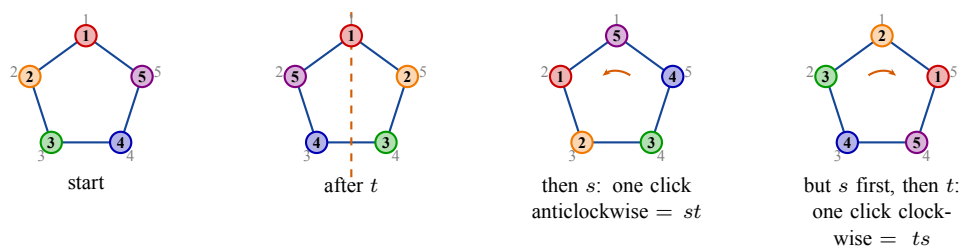


Figure 3.1 Closure in D_5 , and the failure of commutativity. Reflecting in the axis of t and then in the axis of s gives one click of rotation anticlockwise; doing the two reflections in the other order gives one click clockwise. So $st \neq ts$.

Identity and inverses

“Leave the polygon alone” is the rotation by 0° . A rotation is undone by turning back, which is the same as turning forward the rest of the way, so the inverse of one click is four clicks when $n = 5$. A reflection undoes itself: $t^2 = 1$, hence $t^{-1} = t$.

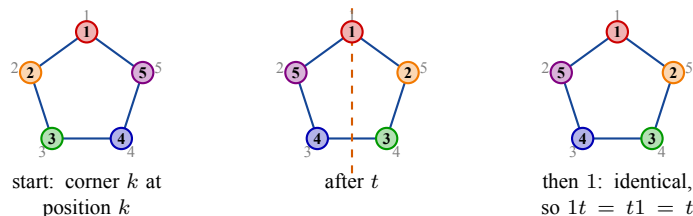


Figure 3.2 Identity in D_5 : start, after the reflection t , then after doing nothing.

Associativity

Free of charge, by the remark on page 11: the elements are functions on the polygon and the rule is composition. So a word like tst has exactly one meaning.

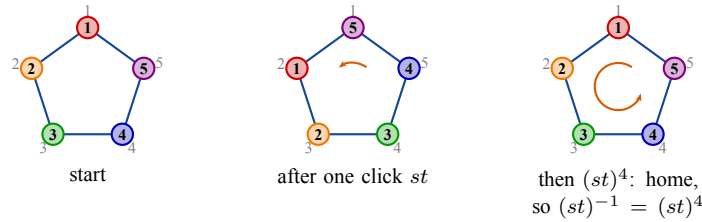
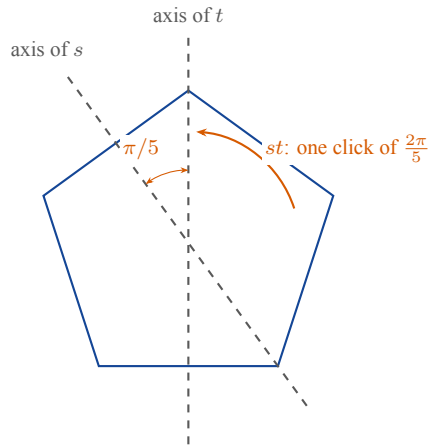


Figure 3.3 Inverses in D_5 : one click of rotation, then four more clicks, returns every corner to its starting position, so $(st)^{-1} = (st)^4$.

The extra properties

$|D_n| = 2n$, and D_n is not abelian for $n \geq 3$. The two reflections s and t drawn in the figures **generate** the group, meaning every element is a product of copies of s and t .



the two reflection axes meet at $\pi/5 = 36^\circ$,
so reflecting in both is a rotation by $2\pi/5 = 72^\circ$

Figure 3.4 The two reflection axes of s and t for the pentagon. They meet at the angle π/n , which is 36° when $n = 5$, and that is what makes the product st a single click of rotation.

Why?

Why st is one click. Reflecting in two lines that meet at angle θ is the same as rotating by 2θ about the point where they cross. The axes of s and t meet at π/n , so st is the rotation by $2\pi/n$: one click. Consequently $(st)^m$ is m clicks, which gives all n rotations as m runs from 0 to $n - 1$, and $(st)^n = 1$.

Why?

Why every reflection is $s(st)^m$. A reflection followed by a rotation is again a reflection, in a different axis; and since $s \cdot s = 1$, the word $s(st)^m$ can be rewritten as $t(st)^{m-1}$. Running m through $0, \dots, n - 1$ produces n distinct reflections, whose axes are π/n apart. Together with the rotations this is $2n$ elements, which is all of them.

Here is the whole group D_5 written out, both as pictures and as words.

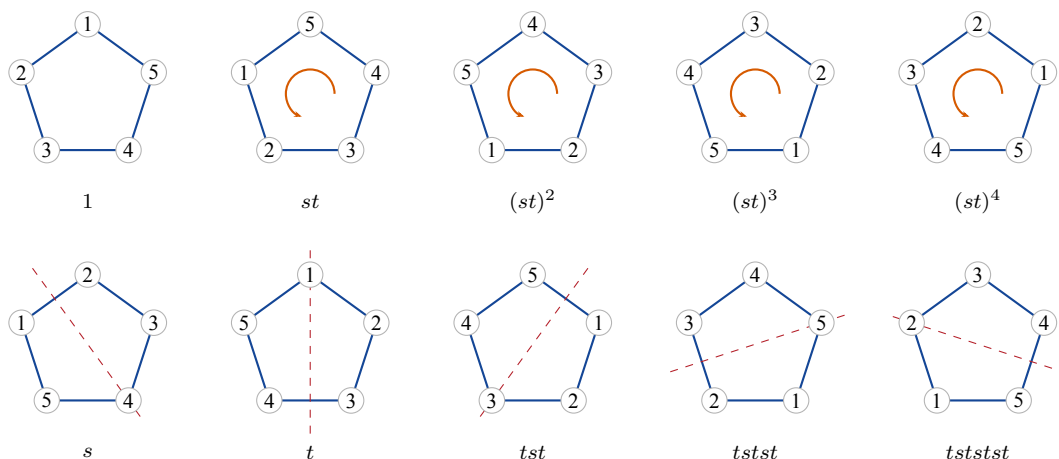


Figure 3.5 The ten symmetries of a regular pentagon. Top row: the five rotations, $(st)^m$ being m clicks of 72° . Bottom row: the five reflections, with their axes dashed. Each circled number shows where that corner has been sent.

word in s, t	type	what it does	corners 1 2 3 4 5 go to
1	rotation	by 0° (0 clicks)	1 2 3 4 5
st	rotation	by 72° (1 click)	2 3 4 5 1
$(st)^2$	rotation	by 144° (2 clicks)	3 4 5 1 2
$(st)^3$	rotation	by 216° (3 clicks)	4 5 1 2 3
$(st)^4$	rotation	by 288° (4 clicks)	5 1 2 3 4
s	reflection	axis at 126° from horizontal	2 1 5 4 3
t	reflection	axis at 90° from horizontal	1 5 4 3 2
tst	reflection	axis at 54° from horizontal	5 4 3 2 1
$tstst$	reflection	axis at 18° from horizontal	4 3 2 1 5
$tststst$	reflection	axis at -18° from horizontal	3 2 1 5 4

Watch out

“ s and t generate D_n ” does not mean every element *is* s or t . It means every element is a product of copies of them. Inverses need not be mentioned here because $s^{-1} = s$ and $t^{-1} = t$, but in general a generating set is allowed to use inverses as well.

The rotations of D_n , taken alone, are closed under the rule and form a group in their own right: a **subgroup** of D_n , of size n . The reflections alone are not, since the product of two of them is a rotation. Finally, the three facts $s^2 = 1$, $t^2 = 1$ and $(st)^n = 1$ turn out to determine D_n completely, in a sense made precise in Section 7.

Try it

Multiply two rows of the table by composing their corner maps, right-hand factor first. For example $t \cdot s$: first s sends 1 2 3 4 5 to 2 1 5 4 3, then t acts on the result. You should land on $(st)^4$, the rotation by -72° , confirming $ts = (st)^{-1}$. Then find which word is the reflection whose axis passes through corner 3.

3.2 Symmetric groups: rearranging n things

The set. All bijections from $\{1, 2, \dots, n\}$ to itself. Such a bijection is called a **permutation**, and the set of them is written S_n .

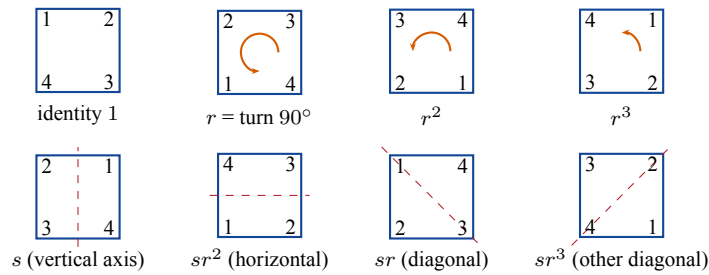


Figure 3.6 The eight symmetries of a square, which is D_4 . Corners are numbered 1, 2, 3, 4 and each picture shows where the numbers end up; r is a quarter turn and s a reflection, and the other three reflections are sr , sr^2 and sr^3 .

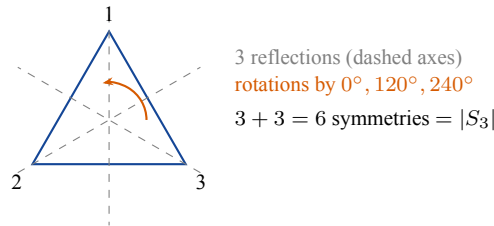


Figure 3.7 The smallest interesting case, D_3 : three rotations and three reflections of an equilateral triangle, six symmetries in all.

The rule. Composition of functions, with the convention of Definition 2.1: in $\tau\sigma$, the permutation σ acts first.

Nothing geometric has to be respected here, which is the point of the example. If you think of $\{1, \dots, n\}$ as n loose points with no distances and no shape, then every rearrangement of them is a symmetry, and S_n is the symmetry group of a set of n points. In the pictures below the points are drawn as n fixed slots, and a coloured disc is the token currently in that slot.

Closure, identity, inverses, associativity

A rearrangement followed by a rearrangement is a rearrangement, since every slot still ends up with exactly one token. The identity is the permutation that fixes every slot, written 1 or $()$. The inverse of a permutation is obtained by sending every token back where it came from, which on the picture means retracing the strands. Associativity is again free, since the rule is composition.

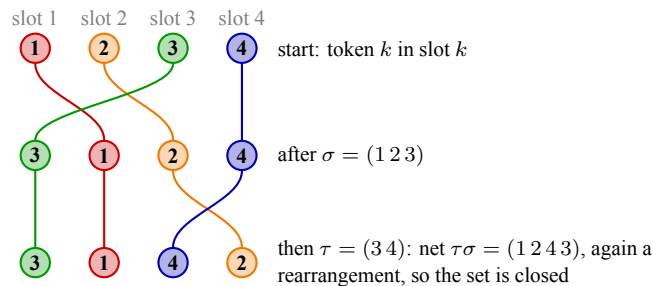


Figure 3.8 Closure in S_4 : after σ , then after τ , the net effect is again a single rearrangement.

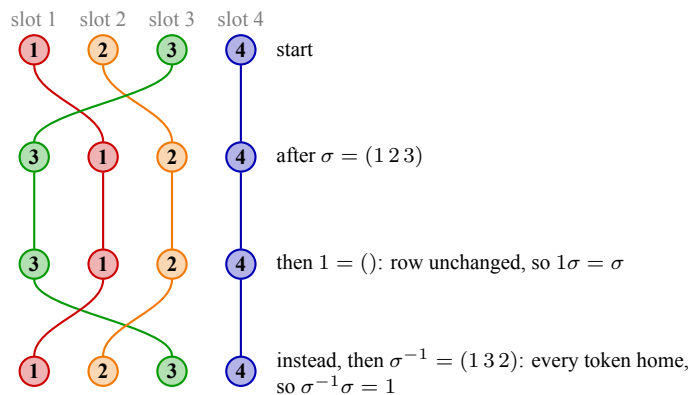


Figure 3.9 Identity and inverses in S_4 in one picture: doing nothing after σ changes no row, and applying σ^{-1} instead brings every token home.

Cycle notation

Writing a permutation as a list of images is correct but unreadable. The standard notation records the *loops* instead.

Definition 3.1. $(i_1\ i_2\ \dots\ i_k)$ denotes the permutation sending i_1 to i_2 , i_2 to i_3 , and so on round to i_k , which goes back to i_1 ; every number not listed is left alone. Such a permutation is a *k-cycle*, and a 2-cycle is called a *transposition* or a *swap*. Permutations written side by side, as in $(1\ 2\ 4)(3\ 5)$, are to be multiplied.

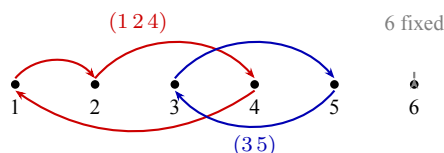


Figure 3.10 The permutation $(1\ 2\ 4)(3\ 5)$ drawn as arrows on the six numbers. Each cycle is a closed loop, disjoint cycles are loops that share no points, and 6 is left alone.

Watch out

The same cycle has several names: $(1\ 2\ 4)$, $(2\ 4\ 1)$ and $(4\ 1\ 2)$ are the same permutation, because the notation lists a loop starting from a chosen point and going round once. The starting point is a choice; the loop is not.

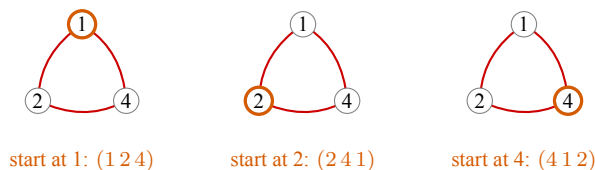


Figure 3.11 One loop, three names.

Multiplying permutations

There is one reliable method: push each number through both factors, remembering that the right-hand factor acts first. Take $(2\ 6) \cdot (1\ 2\ 4)(3\ 5)$.

number	after (1 2 4)(3 5)	then after (2 6)	so the product sends
1	2	6	$1 \rightarrow 6$
2	4	4	$2 \rightarrow 4$
3	5	5	$3 \rightarrow 5$
4	1	1	$4 \rightarrow 1$
5	3	3	$5 \rightarrow 3$
6	6	2	$6 \rightarrow 2$

Now read the answer off as loops. Start at 1 and follow: $1 \rightarrow 6 \rightarrow 2 \rightarrow 4 \rightarrow 1$, which closes up as the 4-cycle (1 6 2 4). The smallest unused number is 3, and $3 \rightarrow 5 \rightarrow 3$ gives (3 5). Everything is accounted for, so

$$(26) \cdot (124)(35) = (1624)(35).$$

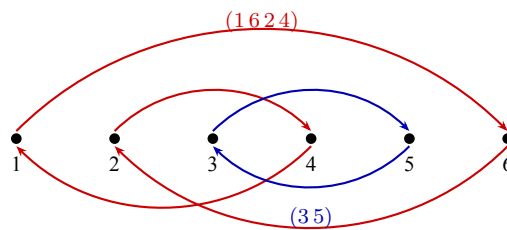


Figure 3.12 The answer as a loop picture.

The same computation can be watched happening to the tokens. Each row is a complete snapshot of where everybody is; the strands between rows show each token's move.

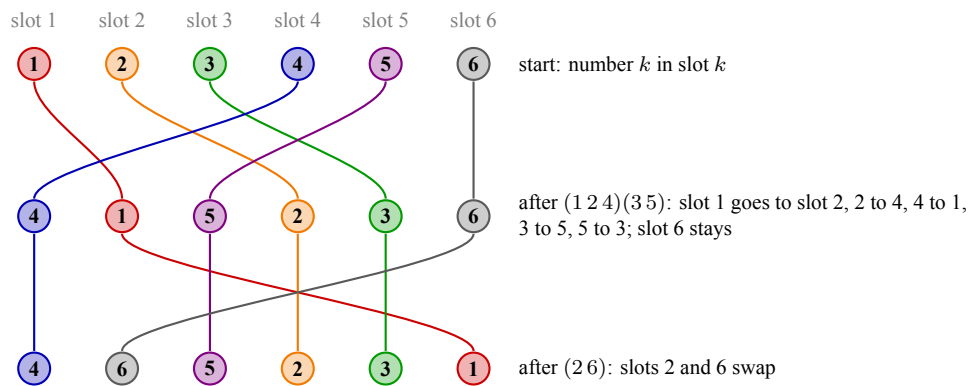


Figure 3.13 One factor at a time. Read the last row: token 1 sits in slot 6, token 6 in slot 2, token 2 in slot 4, token 4 in slot 1, and 3, 5 have swapped. That is the right-hand column of the table above.

Why?

The table and the picture agree because they organise the same information differently. The table follows one *number* at a time through both factors. The picture follows all six at once, one *factor* at a time. Neither is more correct; the table is better for computing, the picture is better for seeing.

Watch out

Apply (2 6) first instead and the first line of the table becomes $1 \rightarrow 1 \rightarrow 2$, and the answer changes to $(1246)(35)$, a different element. The order is a convention, but it is not optional: it is fixed by Definition 2.1.

Permutations as diagrams, and why swaps generate

Draw the slots as n points on a top row and n points on a bottom row, and join each token's start to its finish by a strand. Nudge the picture so that no two crossings happen at the same height. Reading the crossings from top to bottom expresses the permutation as a succession of swaps of *neighbouring* numbers, one per crossing.

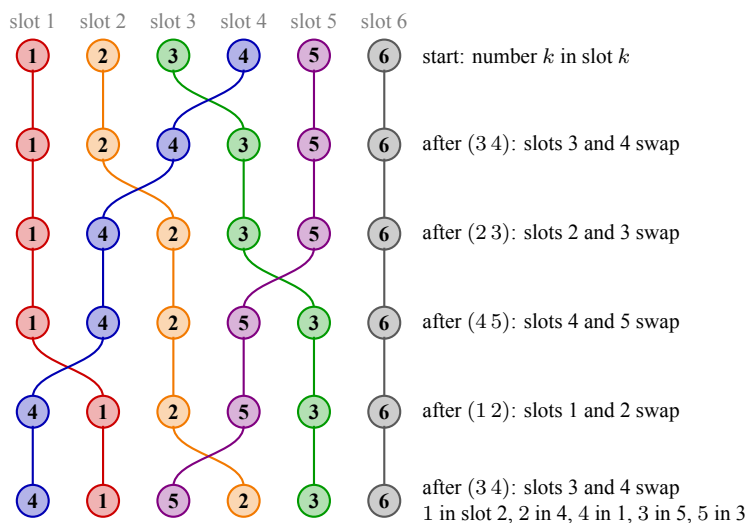


Figure 3.14 The permutation $(1\ 2\ 4)(3\ 5)$ as a stack of five neighbour swaps. One row per height, one swap between consecutive rows.

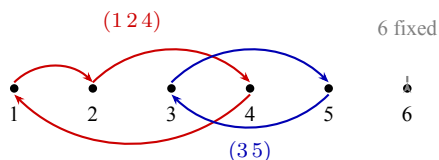


Figure 3.15 The bottom row of that stack, read as loops, is the permutation we started with.

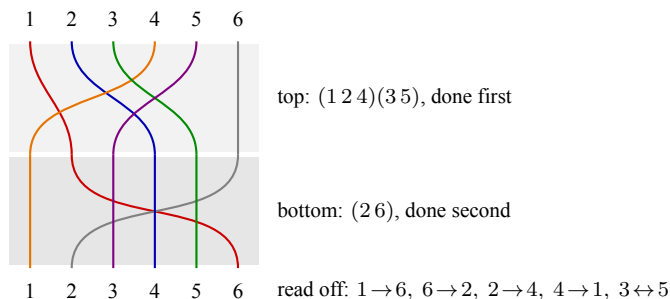


Figure 3.16 Stacking two diagrams multiplies the permutations. The factor performed first goes on top, so the picture reads downwards in the same order as the product $(2\ 6) \cdot (1\ 2\ 4)(3\ 5)$ is computed. Following a strand from the top row to the bottom row gives that number's image under the product.

Proposition 3.2. *Every element of S_n is a product of transpositions; in fact of transpositions of neighbours.*

Proof. Given a permutation, sort the tokens into place by repeatedly swapping a neighbouring pair that is out of order, exactly as in the sorting algorithm the picture above performs. The process ends, because each swap reduces by one the number of pairs that are in the wrong relative order, and that number is finite. Reading the swaps in the order performed expresses the permutation as their product. $\square$

The extra properties

$|S_n| = n!$, because the token in slot 1 has n possible destinations, the next has $n - 1$, and so on. S_n is not abelian once $n \geq 3$: for instance $(1\ 2)(2\ 3) = (1\ 2\ 3)$ while $(2\ 3)(1\ 2) = (1\ 3\ 2)$. By Proposition 3.2 it is generated by neighbour swaps, a generating set of size $n - 1$.

And by Theorem 1.3, S_4 is the rotation group of the cube. With cycle notation available, the dictionary at the end of Section 2 can now be read properly: a quarter turn about a face axis is a 4-cycle such as $(d_1\ d_2\ d_4\ d_3)$, a half turn about a face axis is a pair of swaps such as $(d_1\ d_4)(d_2\ d_3)$, a third of a turn about a diagonal is a 3-cycle such as $(d_2\ d_4\ d_3)$, and a half turn about an edge axis is a single swap such as $(d_1\ d_2)$.

Remark 3.3. The full symmetry group of the cube, reflections included, has 48 elements and is *not* S_4 . It is S_4 paired with a two-element group that records whether a motion is a mirror image. Products of groups are not needed in this chapter, but that is where the extra factor comes from.

Try it

Compute $(1\ 2\ 3\ 4) \cdot (1\ 2\ 3\ 4)$ and $((1\ 2)(3\ 4))^{-1}$. Then write $(1\ 3)(2\ 4)$ as a product of neighbour swaps, and check your answer by composing them.

3.3 The integers modulo n

This is the first group in the gallery that is not presented as the symmetries of something you can hold, and the first whose multiplication is given by a *formula with cases*. Both facts mean the axioms have to be checked by hand, and the third one costs real work.

The set. $\mathbb{Z}/n\mathbb{Z} = \{0, 1, \dots, n - 1\}$, for a fixed integer $n > 1$.

The rule. Add, and if the answer reaches n , subtract n :

$$a \oplus b = \begin{cases} a + b, & a + b \leq n - 1, \\ a + b - n, & a + b \geq n. \end{cases}$$

What the elements really are

Each symbol $0, \dots, n - 1$ is a name for a whole family of integers: those leaving that remainder on division by n . Colour every integer by its remainder and the picture is a repeating pattern; wrap the line round a circle with n points and each colour lands on one point.

Why?

Why the rule needs only two cases. Both a and b are at most $n - 1$, so the ordinary sum $a + b$ is at most $2n - 2$, which is less than $2n$. One subtraction therefore always suffices, and $a \oplus b$ is exactly the remainder of $a + b$ on division by n . This is the reason the operation is called *arithmetic modulo n* and written $a + b \equiv c \pmod{n}$.

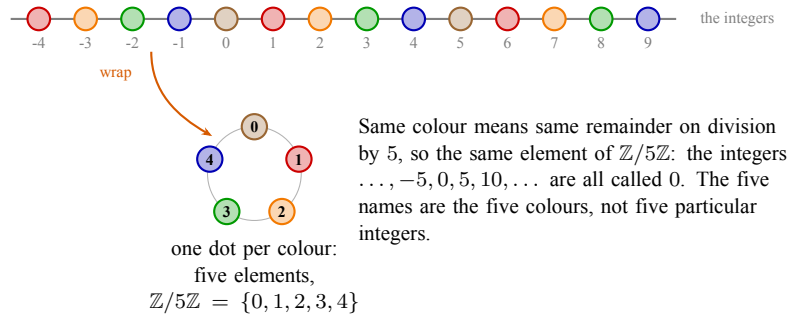


Figure 3.17 Why there are exactly five elements when $n = 5$. Same colour means same remainder, hence the same element of $\mathbb{Z}/5\mathbb{Z}$; the integers $\dots, -5, 0, 5, 10, \dots$ all go by the name 0.

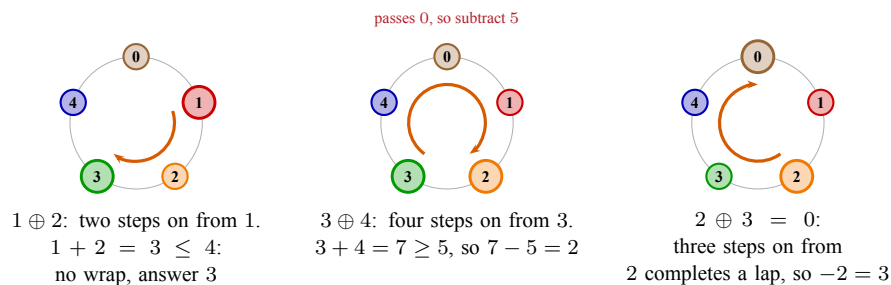


Figure 3.18 The rule on the circle for $n = 5$. Left: no wrap. Middle: the walk passes 0, so n is subtracted. Right: the inverse of 2 is 3, because three more steps complete a lap.

The three axioms

Identity. 0 works: $a + 0 = a \leq n - 1$, so no wrap occurs and $0 \oplus a = a \oplus 0 = a$. On the circle, adding 0 is standing still.

Inverses. The inverse of 0 is 0. For $m \neq 0$ the inverse is $n - m$, because $m + (n - m) = n \geq n$, so the rule subtracts n and the answer is 0. On the circle, walking $n - m$ steps on from m completes exactly one lap.

Associativity. Here the cases bite: each $\oplus$ may or may not subtract n , so comparing $(a \oplus b) \oplus c$ with $a \oplus (b \oplus c)$ seems to need a list of combinations. A worked case with $n = 5, a = 3, b = 4, c = 2$:

$(a \oplus b) \oplus c$	$a \oplus (b \oplus c)$
$3 \oplus 4 = 7 - 5 = 2$	$4 \oplus 2 = 6 - 5 = 1$
$2 \oplus 2 = 4$, no wrap	$3 \oplus 1 = 4$, no wrap
both equal 4, the remainder of $3 + 4 + 2 = 9$ on division by 5	

Why?

The argument, with no cases at all. Each application of $\oplus$ either keeps the sum or subtracts n . So however you bracket, the answer is the integer $a + b + c$ minus some multiple of n ; and it lies in $\{0, \dots, n - 1\}$, because that is where $\oplus$ always lands. But there is only *one* integer in $\{0, \dots, n - 1\}$ differing from $a + b + c$ by a multiple of n , namely its remainder. Both bracketings therefore equal that remainder, hence each other. In one sentence: $\oplus$ is “add in $\mathbb{Z}$, then take the remainder”, and taking remainders is compatible with addition.

Watch out

The word “multiplication” in Definition 2.2 here means *addition*. Ordinary multiplication modulo n is not a group rule on this set, because 0 has no multiplicative inverse. Note also that $\mathbb{Z}/n\mathbb{Z}$ is not a subset of $\mathbb{Z}$ in any useful sense: $3 \oplus 4 = 2$ in $\mathbb{Z}/5\mathbb{Z}$ while $3 + 4 = 7$ in $\mathbb{Z}$.

The extra properties

$|\mathbb{Z}/n\mathbb{Z}| = n$, and the group is **abelian**, since $a + b = b + a$ in $\mathbb{Z}$ and the wrap does not care about order. It is **cyclic**: starting at 0 and adding 1 repeatedly visits every element, so $\{1\}$ generates it all by itself. More generally m generates $\mathbb{Z}/n\mathbb{Z}$ exactly when m and n have no common factor.

Why?

Why the condition is $\gcd(m, n) = 1$. If m and n have no common factor, there are integers u, v with $um + vn = 1$; reducing modulo n gives $u \cdot m \equiv 1$, so adding m to itself u times produces 1, and 1 generates everything. Conversely, if $d > 1$ divides both m and n , then every multiple of m is again a multiple of d modulo n , so 1 is never reached.

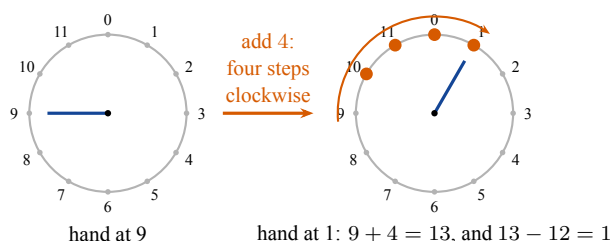


Figure 3.19 Familiar instances: the 12-hour clock is $\mathbb{Z}/12\mathbb{Z}$, and $9 + 4 = 13$ is renamed 1.

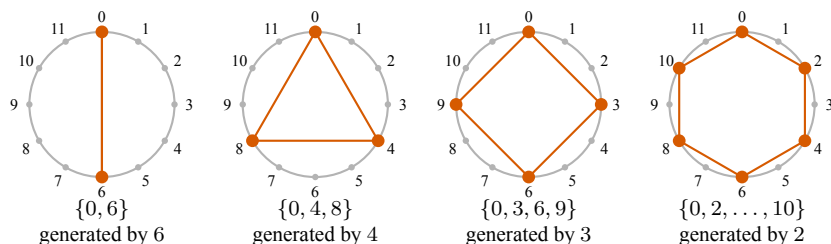


Figure 3.20 Subgroups of $\mathbb{Z}/12\mathbb{Z}$. Each highlighted set is closed under “add and wrap”, so each is a group in its own right, and each is a regular polygon inscribed in the clock, one for every divisor of 12. Since 5 and 12 have no common factor, adding 5 repeatedly visits all twelve points; starting from 4 you only ever reach the triangle.

Try it

In $\mathbb{Z}/12\mathbb{Z}$: what is the inverse of 5? of 8? which elements are their own inverses? Check associativity for $a = 7$, $b = 9$, $c = 8$ by both bracketings, and against the remainder of 24. Then say which axiom fails for the rule “subtract, then wrap”.

3.4 Decorating a polygon to see $\mathbb{Z}/n\mathbb{Z}$

$\mathbb{Z}/n\mathbb{Z}$ was defined by a formula, but it is a symmetry group after all, and the object it is the symmetry group of is obtained from the n -gon by a trick worth learning, because the same trick will be used three more times.

The symmetries of a regular n -gon form D_n , which is twice too big. So spoil the polygon: draw a little

arrow on each edge, all running the same way round. A rotation carries arrows to arrows pointing the same way round and is still a symmetry of the decorated polygon. A reflection reverses every arrow, and is not. The decoration has removed exactly the reflections.

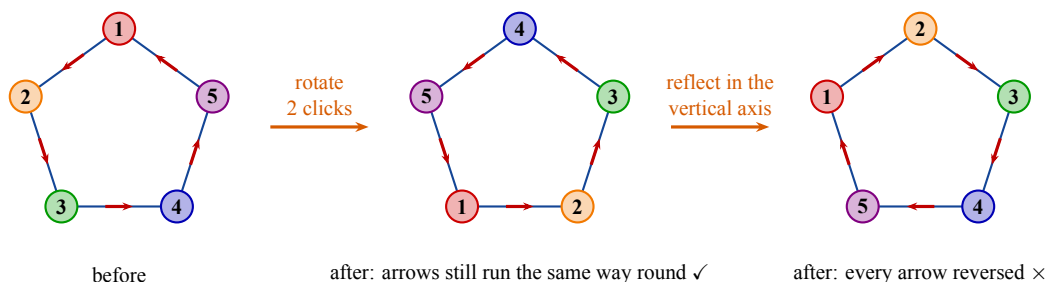


Figure 3.21 A rotation preserves the arrows; a reflection reverses them. So the symmetry group of the arrowed n -gon is the n rotations, and the rotations compose by adding clicks modulo n .

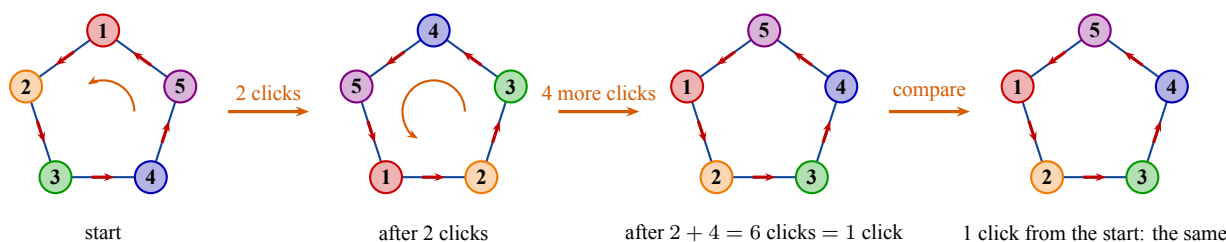


Figure 3.22 Composing clicks is addition modulo n : two clicks then four clicks is six clicks, and six clicks of a pentagon is one click, because five clicks is a full turn. In symbols $2 + 4 \equiv 1 \pmod{5}$.

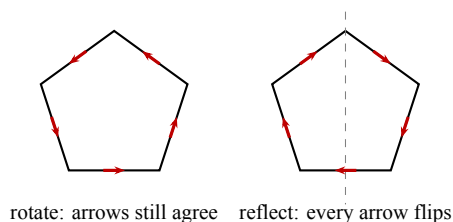


Figure 3.23 The arrowed n -gon, whose symmetry group is $\mathbb{Z}/n\mathbb{Z}$. It is also the Cayley graph of $\mathbb{Z}/n\mathbb{Z}$ with the generator 1, in the sense of Section 4.

Big picture

This is the first example of the chapter's main theme in miniature. An algebraic object, $\mathbb{Z}/n\mathbb{Z}$, was matched with a geometric one, the arrowed polygon, and the matching turned addition with wrap into rotation. Section 4 shows this is not a lucky accident: *every* group can be presented this way, once you know how to build the right decorated object.

3.5 The integers as slides of a line

The set. $\mathbb{Z}$. *The rule.* Addition. Identity 0, inverse $-n$, associativity inherited from ordinary arithmetic. So $\mathbb{Z}$ is a group, and an abelian one.

To see it as a symmetry group, mark the integer points on a line. A rigid motion of the line that sends marked points to marked points is either a slide by a whole number of units, or a reflection about a marked

point, or a reflection about a midpoint between two neighbours. That collection is bigger than $\mathbb{Z}$; it is called the *infinite dihedral group* D_∞ , the $n \rightarrow \infty$ version of D_n .

Why?

Why the slide passes both tests and the reflection does not. Sliding by 3 is the map $x \mapsto x + 3$, the same shift at every point, so gaps are preserved, $|(x+3) - (y+3)| = |x - y|$, and order is preserved, $x < y \Rightarrow x+3 < y+3$. An arrow records only “ k is to the left of $k+1$ ”, so it keeps pointing right. Reflecting about the point 2 is $x \mapsto 4 - x$: gaps are preserved too, but now $x < y$ becomes $4 - x > 4 - y$, so every arrow turns round.

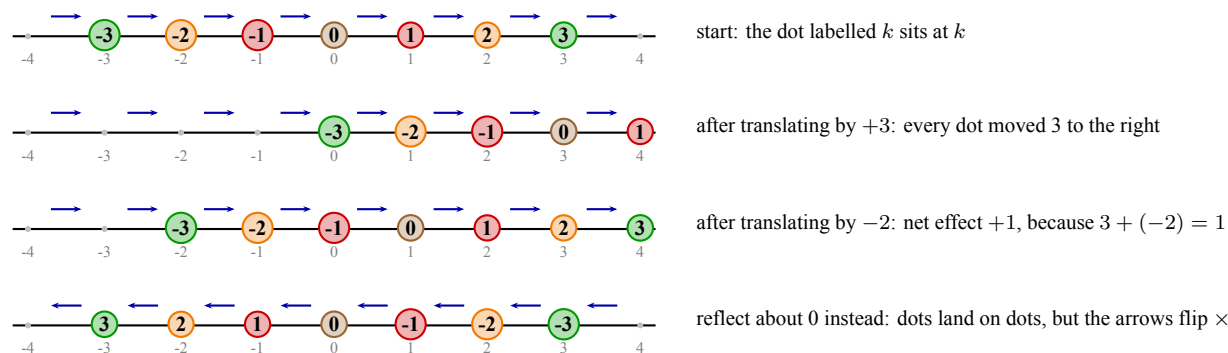
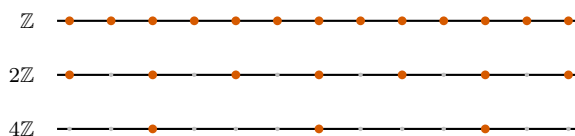


Figure 3.24 Composing slides is adding integers: slide by $+3$, then by -2 , and the net effect is $+1$. The arrowed line is the Cayley graph of $\mathbb{Z}$.



a chain of subgroups $\mathbb{Z} \supset 2\mathbb{Z} \supset 4\mathbb{Z}$; each is a copy of $\mathbb{Z}$ (multiplication by 2)

Figure 3.25 Subgroups of $\mathbb{Z}$. Every subgroup is the set of multiples of some $m \geq 0$. The map $n \mapsto 2n$ shows $\mathbb{Z}$ containing a smaller copy of itself, something no finite group can do.

3.6 The plane lattice $\mathbb{Z}^2$

The set. $\mathbb{Z}^2 = \{(m, n) : m, n \in \mathbb{Z}\}$, pairs of integers. *The rule.* Add coordinates: $(m, n) + (m', n') = (m + m', n + n')$. Identity $(0, 0)$, inverse $(-m, -n)$, associativity inherited coordinatewise. Abelian.

Geometrically, (m, n) is the slide “ m to the right and n up”, and composing two slides adds the coordinates, which is vector addition.

The object whose symmetries are exactly $\mathbb{Z}^2$ is the integer grid, decorated so that rotations and reflections are spoiled: colour the horizontal edges one colour and the vertical edges another.

Watch out

$\mathbb{Z}^2$ is abelian, and it is generated by two elements, $(1, 0)$ and $(0, 1)$. In Section 3.8 we meet another group with two generators which is as far from abelian as possible. Two generators say very little about a group on their own.

3.7 Matrix groups, in one paragraph

For completeness, one important family that is not a symmetry group of a polygon or a cube. Let $\text{GL}(n, \mathbb{R})$ be the set of $n \times n$ matrices with real entries and nonzero determinant, with matrix multiplication as the

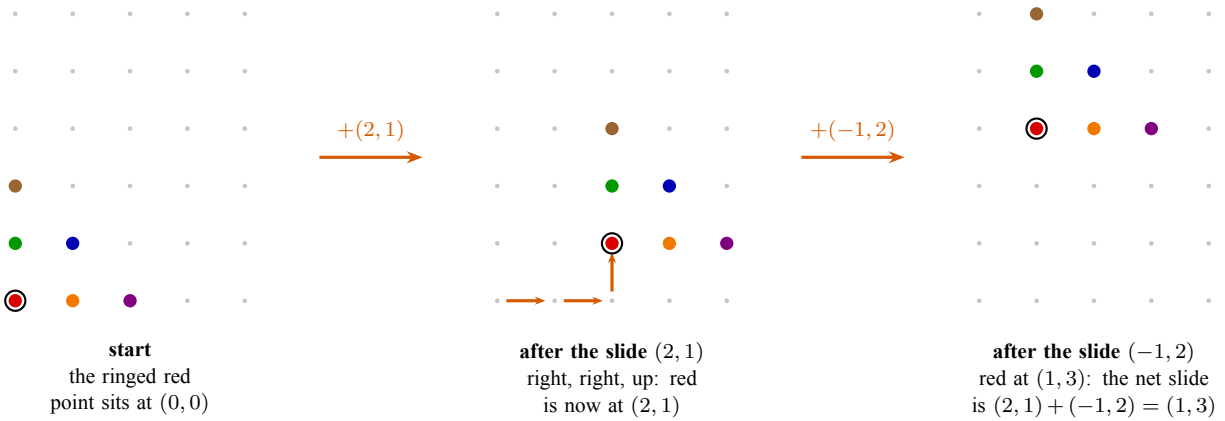


Figure 3.26 Slides of the plane compose by adding coordinates. Every point of the lattice moves by the same amount, so it is enough to follow one of them: the ringed red point. The middle panel also breaks the slide $(2,1)$ into unit steps right, right and up, so $(2,1) = 2(1,0) + 1(0,1)$ and the two unit slides generate every slide.

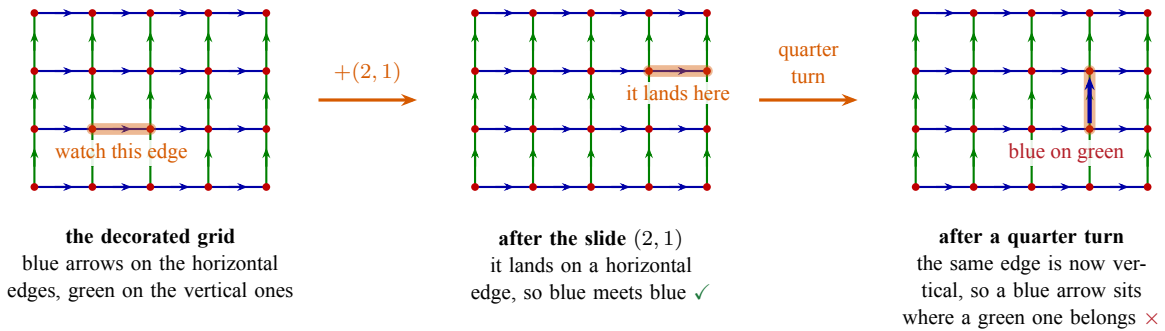


Figure 3.27 Which motions of the plane survive the decoration. The background grid is the same in all three panels: it is the object. Follow the highlighted edge. A slide carries it to another horizontal edge, so blue still meets blue and the slide is a symmetry. A quarter turn carries it to a vertical edge, where a blue arrow would have to sit on a green one, so the quarter turn is not. A reflection fails for a different reason: it reverses the arrows. Only the slides survive, and they compose exactly like $\mathbb{Z}^2$.

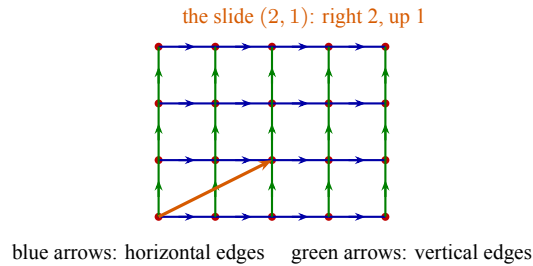


Figure 3.28 The decorated grid, whose symmetry group is $\mathbb{Z}^2$, and which is also its Cayley graph with the two unit slides as generators.

rule. Closure holds because $\det(AB) = \det(A)\det(B)$ is nonzero when both factors are; the identity is the identity matrix; inverses exist exactly because the determinant is nonzero; and associativity holds because matrix multiplication represents composition of linear maps, which is composition of functions. Requiring determinant exactly 1 gives a subgroup $\text{SL}(n, \mathbb{R})$. Replacing $\mathbb{R}$ by $\mathbb{Z}/p\mathbb{Z}$ for a prime p gives

finite examples: the invertible 2×2 matrices over $\mathbb{Z}/2\mathbb{Z}$ form a group with 6 elements which behaves exactly like S_3 , since it permutes the three nonzero vectors of the plane over $\mathbb{Z}/2\mathbb{Z}$.

3.8 The free group of rank two

The last and most important example in the gallery. Every earlier group arrived with its elements already in hand: rotations, permutations, remainders, pairs of integers. Here the *set* is defined by a condition and the *rule* is a two-step recipe, so even closure has to be argued, and associativity becomes a genuine theorem.

The alphabet. Four symbols: a, b, a^{-1}, b^{-1} . At this stage a^{-1} is nothing but a symbol; the notation is chosen because it will turn out to be an inverse once the group exists.

Words. A **word** is any finite string of these four symbols, including the **empty word**, which has no letters at all. A word is **reduced** if it never has a symbol standing next to its partner, that is, no aa^{-1} , $a^{-1}a$, bb^{-1} or $b^{-1}b$ anywhere in it.

The set. $F_2 = \{\text{reduced words in } a, b\}$.

The rule. Write one word after the other, then delete forbidden pairs until none is left.

In the figures each letter is a box, red for the a family and blue for the b family, and a shaded pair is the one about to be deleted.

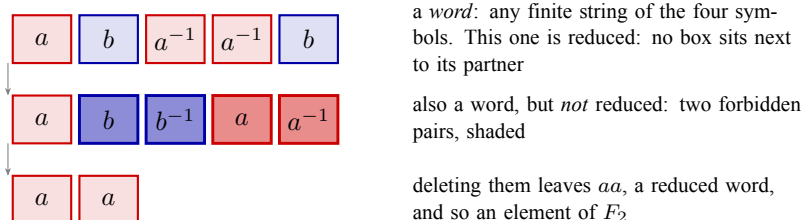


Figure 3.29 A word, a word that is not reduced, and the reduced word left after deleting.

Closure

Two things need checking here, and both are easy to overlook. First, the deleting *stops*: each deletion removes two boxes, so a string of length k can be cut at most $k/2$ times. Second, what it stops at is a *reduced* word, because that is precisely the condition for no deletion being available. So the answer lands back in F_2 .

There is also a useful picture of where the cancelling happens. Both factors were already reduced, so the only forbidden pair in the concatenation is at the *seam*, and the deleting unzips outwards from there.

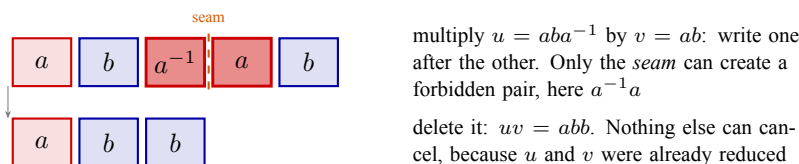


Figure 3.30 Multiplying aba^{-1} by ab : only the seam can create a forbidden pair.

Identity

The empty word. Writing nothing in front of or behind a word w changes no letter and creates no forbidden pair, so $1w = w1 = w$.

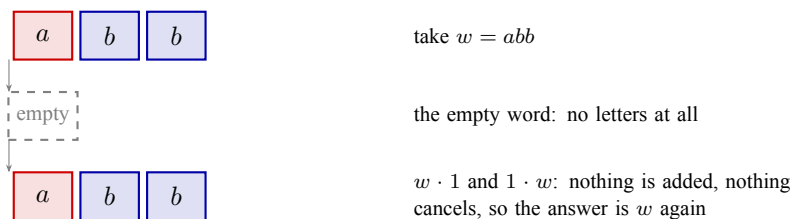


Figure 3.31 The identity of F_2 is the word with no letters.

Inverses

Reverse the word and flip every letter, turning a into a^{-1} , a^{-1} into a , and likewise for b . This is the socks-and-shoes rule again: to undo “socks, then shoes” you take the shoes off first. The two strings then annihilate from the middle outwards.

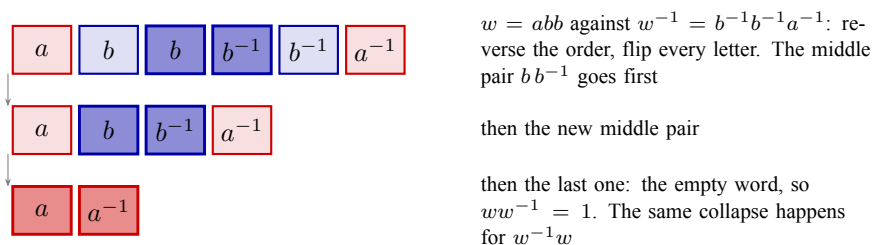


Figure 3.32 $w = abb$ against $w^{-1} = b^{-1}b^{-1}a^{-1}$: the middle pair goes first, then the next, then the last, leaving the empty word.

Associativity

This is the axiom that costs something. $(uv)w$ and $u(vw)$ are two different *orders of deleting* inside the one long string uvw , and there is no reason on the face of it why different orders should reach the same reduced word. Take $u = ab$, $v = b^{-1}a^{-1}$, $w = ab$.

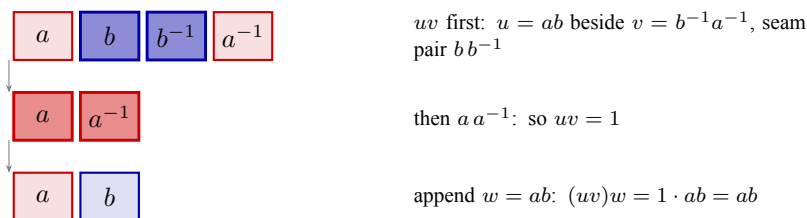
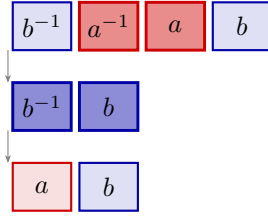


Figure 3.33 $(uv)w$: the product uv collapses completely, and the answer is ab .

Theorem 3.4. Every word in a, b, a^{-1}, b^{-1} reduces to exactly one reduced word, whatever order the deletions are performed in.

Proof sketch. Induct on the length of the word. If two deletions are available at once, they either involve four distinct positions, in which case performing one leaves the other still available and the two orders lead



vw first: $v = b^{-1}a^{-1}$ beside $w = ab$, seam pair $a^{-1}a$

then $b^{-1}b$: so $vw = 1$

put $u = ab$ in front: $u(vw) = ab \cdot 1 = ab$.
Same answer

Figure 3.34 $u(vw)$: now vw collapses completely, and the answer is again ab . The two routes deleted different pairs in a different order.

to the same shorter word; or they overlap in a position, in which case the overlap forces a pattern such as $xx^{-1}x$, and deleting either pair leaves the same single letter x . Either way any two first steps can be followed by further deletions reaching a common word, and the induction hypothesis applies to it. $\square$

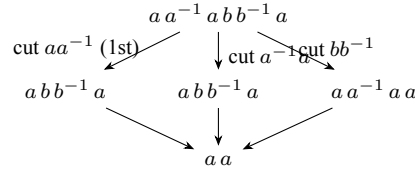
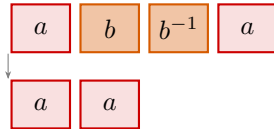


Figure 3.35 The word $aa^{-1}abb^{-1}a$ has three places where a deletion is available. Whichever you choose, and whatever you do next, the process ends at the same reduced word aa .

Why?

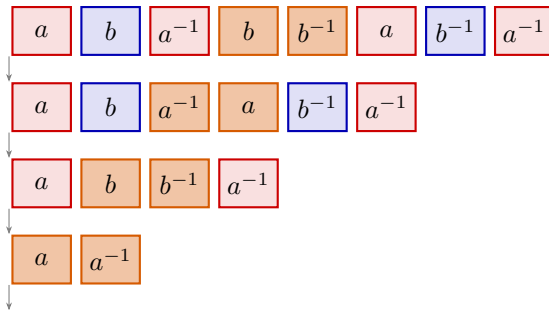
Why Theorem 3.4 gives associativity. Reduce the single string uvw in any order at all. Both $(uv)w$ and $u(vw)$ are such reductions, since each is obtained by deleting pairs inside uvw , merely in a prescribed order. By the theorem they end at the same reduced word, so they are equal, which is (G3). *This is the price of defining a group by strings instead of by functions:* for the cube, for S_n and for D_n associativity came free from composition of functions.



concatenate ab and $b^{-1}a$: the pair bb^{-1} is not allowed

delete it: the product is aa

Figure 3.36 Multiplying in F_2 : concatenate, then reduce.



$aba^{-1}b$ followed by $b^{-1}ab^{-1}a^{-1}$:
cancel bb^{-1} in the middle

cancel $a^{-1}a$

cancel bb^{-1}

cancel aa^{-1}

nothing left: the empty word, the identity

Figure 3.37 Checking an inverse: $aba^{-1}b$ against $b^{-1}ab^{-1}a^{-1}$ collapses from the middle outwards to the empty word.

The extra properties

$\{a, b\}$ generates F_2 by construction, since every element is a word in a and b and their inverses. The group is infinite; more precisely there are $4 \cdot 3^{k-1}$ reduced words of length k , because after the first letter each new letter may be anything except the partner of the one before, so the number of elements grows exponentially with length.

It is not abelian: ab and ba are different reduced words, hence different elements. In fact no element except the identity has finite order, since a nonempty reduced word never collapses when repeated.

The word **free** means precisely that the only relations holding in F_2 are the forced ones $xx^{-1} = 1$. Nothing else is ever true there. Two consequences: when you build a homomorphism out of F_2 you may send a and b absolutely anywhere (Section 5), and every group whatever can be obtained from a free group by imposing extra relations (Section 7).

Watch out

Compare $\mathbb{Z}^2$ of Section 3.6: the same two generators, one extra rule $ab = ba$, and a completely different group. In $\mathbb{Z}^2$ there are about k^2 elements of length k ; in F_2 there are about 3^k . Adding one relation changed polynomial growth into exponential growth.

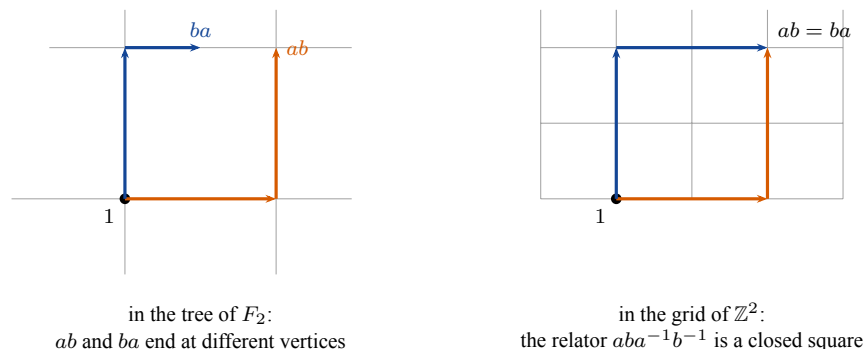


Figure 3.38 The words ab and ba as paths, and why they end at different places.

Try it

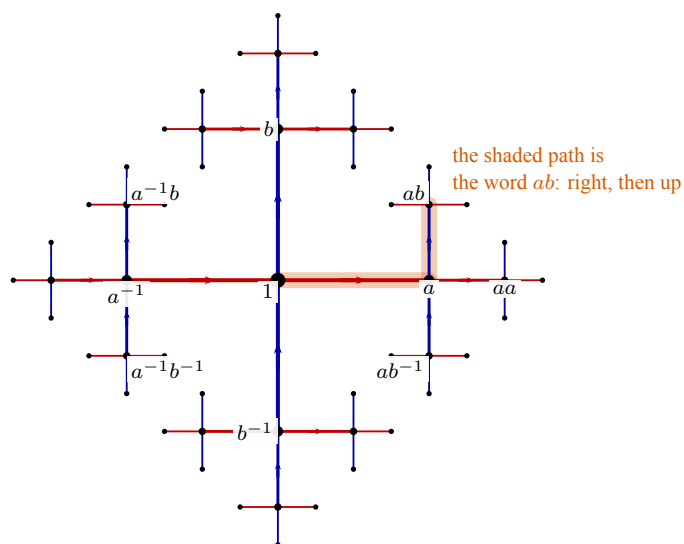
Reduce $baa^{-1}b^{-1}ba$ in two different orders and confirm that both give ba . Write down the inverse of $a^{-1}bab^{-1}$ and check the collapse. How many reduced words have length 3? Finally, decide which axiom would fail if the set were taken to be *all* words, not just the reduced ones, with plain concatenation as the rule.

4 Generators and the Cayley graph

Three of the examples above were built by the same trick: take a geometric object, decorate it so that unwanted symmetries are spoiled, and read off the group that survives. This section turns the trick round. Instead of starting with an object and finding its group, start with a group and *build* an object.

4.1 Generating sets

Definition 4.1. A subset S of a group G **generates** G if every element of G can be written as a product of finitely many elements of S and inverses of elements of S . Such a product is called a **word** in S , and S is a **generating set**.



red edges: multiply by a , arrowheads pointing rightwards
blue edges: multiply by b , arrowheads pointing upwards

Figure 3.39 The object whose symmetry group is F_2 , drawn the same way as every other picture in this chapter: one dot per group element, and one coloured arrow per generator. Start at the dot marked 1 and read a word left to right, taking a red edge rightwards for a and backwards along a red edge for a^{-1} , a blue edge upwards for b and backwards for b^{-1} . The shaded path is the word ab : right, then up. Edges are drawn shorter at each step outwards, so that the whole tree fits; every dot really has four edges at it. Because a reduced word never doubles back, distinct reduced words end at distinct dots, which is why F_2 is infinite and why the picture is a tree rather than a grid.

Every example so far came with one.

group	a generating set	typical element as a word
$\mathbb{Z}$	$\{1\}$	$5 = 1 + 1 + 1 + 1 + 1$
$\mathbb{Z}/n\mathbb{Z}$	$\{1\}$	every element, by repeated addition
$\mathbb{Z}^2$	$\{(1, 0), (0, 1)\}$	$(2, 1) = \text{right, right, up}$
D_n	$\{s, t\}$, two reflections	$(st)^m$ and $s(st)^m$
S_n	the $n - 1$ neighbour swaps	any permutation, by sorting
rotations of the cube	two quarter turns about different axes	every one of the 24
F_2	$\{a, b\}$	the word itself

A group can have many generating sets of very different sizes, and finding a small one is often the first step in understanding it. Note that S generating G says nothing about *uniqueness*: in $\mathbb{Z}$ the element 0 is both the empty product and $1 + (-1)$.

Try it

Show that $\{2, 3\}$ generates $\mathbb{Z}$ but $\{2\}$ does not. Show that $\{4\}$ does not generate $\mathbb{Z}/12\mathbb{Z}$ but $\{5\}$ does. Then convince yourself that the three face half turns of the cube together with the identity form a group that no single element generates.

4.2 The Cayley graph

Definition 4.2. Let G be a group and S a generating set. The **Cayley graph** of G with respect to S is the following decorated object. Put one dot for each element of G . For each element g and each generator $s \in S$, draw an arrow from the dot g to the dot gs , coloured according to which generator s it came from.

Three features of the definition deserve a sentence each. The dots are the group elements, so the graph is as big as the group. The arrows are labelled by generators, so the same picture with a different generating set looks different. And the arrows are coloured and directed, which is exactly the decoration that spoils unwanted symmetries, as in Sections 3.4 and 3.6.

Every picture in the list below has already appeared in this chapter, without the name.

group	generating set	Cayley graph	drawn in
$\mathbb{Z}/n\mathbb{Z}$	$\{1\}$	the arrowed n -gon	Figure 3.23
$\mathbb{Z}$	$\{1\}$	the arrowed line	Figure 3.24
$\mathbb{Z}^2$	$\{(1, 0), (0, 1)\}$	the grid with coloured edges	Figure 3.28
F_2	$\{a, b\}$	the 4-valent tree	Figure 3.39
D_n	$\{s, t\}$	a $2n$ -gon with two edge colours	Problem 10

Why?

Why the arrow goes from g to gs and not to sg . Because of the convention of Definition 2.1. Walking along an s -arrow means multiplying on the *right* by s . Reading a path from the base dot 1 then spells out a word in the generators from left to right, which is exactly how words are written. If the arrows ran to sg instead, reading a path would spell words backwards.

row · column:

	1	r	r^2	s	sr	sr^2
1	1	r	r^2	s	sr	sr^2
r	r	r^2	1	sr^2	s	sr
r^2	r^2	1	r	sr	sr^2	s
s	s	sr	sr^2	1	r	r^2
sr	sr	sr^2	s	r^2	1	r
sr^2	sr^2	s	sr	r	r^2	1

Figure 4.1 A group's multiplication table and its Cayley graph carry the same information. The table is what you compute with; the graph is what you can look at.

Big picture

The punchline of the office hour. Take a group G , choose a generating set, and build the Cayley graph. That graph is a decorated object, and one can ask for its symmetries: the motions of the graph sending dots to dots, arrows to arrows, and respecting the colours. The answer is that these symmetries are exactly the elements of G again, with g acting by sending each dot x to the dot gx . So *every group is the symmetry group of some object*, and the object can be manufactured from the group itself. The formal statement and proof belong to the next office hour; what matters here is that the slogan “groups are symmetries” is a theorem, not a slogan.

Try it

Draw the Cayley graph of $\mathbb{Z}/6\mathbb{Z}$ with the generating set $\{1\}$, and then with $\{2, 3\}$. The two pictures look nothing alike, yet they describe the same group. Then draw the Cayley graph of S_3 with its two neighbour swaps; you should get a hexagon with two colours of edge.

5 Comparing groups: homomorphisms

Two groups can be built from completely different material and still behave identically, and a group can contain a smaller group inside it, or shadow a smaller one. The language for all three situations is the same.

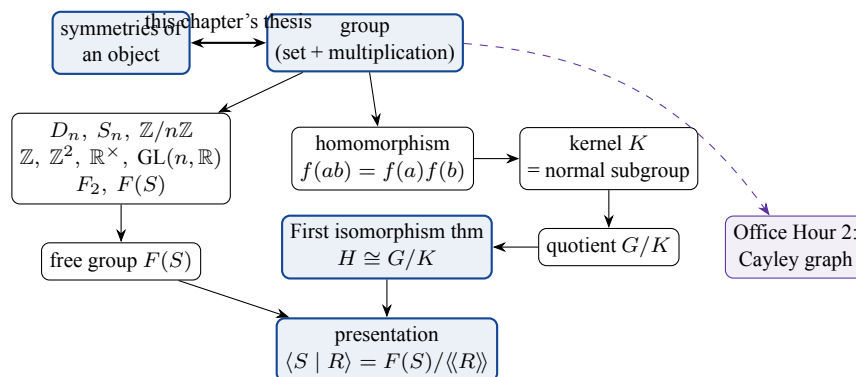


Figure 5.1 The three notions this chapter connects, and the arrows between them. A group is a set with a multiplication; a symmetry group is the collection of motions preserving an object; and a homomorphism is a map between groups that respects multiplication.

Definition 5.1. Let G and H be groups. A function $f : G \rightarrow H$ is a **homomorphism** if

$$f(gh) = f(g)f(h) \quad \text{for all } g, h \in G.$$

It is an **isomorphism** if in addition it is a bijection, and then G and H are called **isomorphic**, written $G \cong H$.

The defining equation deserves to be read slowly, because the two multiplications in it are different. On the left, gh is computed in G ; on the right, $f(g)f(h)$ is computed in H . The condition says these two ways of combining give the same answer: *multiply first and then translate, or translate first and then multiply*.

$$\begin{array}{ccc} (a, b) \in G \times G & \xrightarrow{f \text{ on each}} & (f(a), f(b)) \in H \times H \\ \text{multiply in } G \downarrow & & \downarrow \text{multiply in } H \\ ab \in G & \xrightarrow{f} & f(a)f(b) = f(ab) \in H \end{array}$$

Figure 5.2 The two routes a homomorphism must make agree.

Example 5.2. $f : \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ sending even numbers to 0 and odd numbers to 1. Adding two integers and then taking parity gives the same answer as taking parities and then adding modulo 2: odd plus odd is even, and $1 + 1 = 0$.

Example 5.3. $\text{sign} : S_n \rightarrow \mathbb{Z}/2\mathbb{Z}$, sending a permutation to 0 if it is a product of an even number of swaps and to 1 if odd. That this is well defined is a theorem, and the parity of the number of swaps can be computed as the parity of the number of crossings in a strand diagram, or of the number of pairs whose order is reversed.

Example 5.4. $n \mapsto 2n$ is an injective homomorphism $\mathbb{Z} \rightarrow \mathbb{Z}$ which is not onto; its image is the subgroup of even integers. So $\mathbb{Z}$ contains a copy of itself as a proper subgroup, which is impossible for a finite group.

in $\mathbb{Z}$	$\xrightarrow{\text{parity}}$	in $\mathbb{Z}/2\mathbb{Z}$
$7 + 5 = 12$	$\xrightarrow{\text{parity}}$	$1 + 1 = 0 \checkmark$
$7 \cdot (-3) = -21$	$\xrightarrow{\text{parity}}$	$1 \cdot 1 = 1 \checkmark$
$4 + 6 = 10$	$\xrightarrow{\text{parity}}$	$0 + 0 = 0 \checkmark$

Figure 5.3 Parity as a homomorphism $\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$: add first and take parity, or take parities and add in $\mathbb{Z}/2\mathbb{Z}$.

$(2\ 6)$: one swap, so **odd**

$(1\ 2\ 4)(3\ 5) = (3\ 4)(1\ 2)(4\ 5)(2\ 3)(3\ 4)$: five swaps, so **odd**

product $(1\ 6\ 2\ 4)(3\ 5)$: a 4-cycle is 3 swaps, a 2-cycle is 1 swap, total 4, so **even**

odd $\cdot$ odd = even, i.e. $1 + 1 = 0$ in $\mathbb{Z}/2\mathbb{Z}$: the sign respects multiplication $\checkmark$

Figure 5.4 The sign homomorphism on the product computed in Section 3.2, counted by crossings.

Example 5.5. The bijection of Theorem 1.3 is an isomorphism from the rotation group of the cube to S_4 : performing two rotations one after another corresponds to composing the two rearrangements of the diagonals. This is the sentence that makes the theorem useful rather than merely a count.

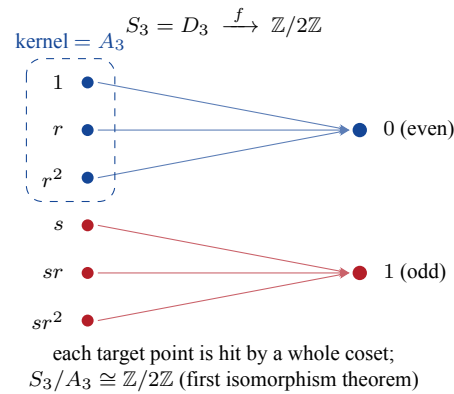


Figure 5.5 A homomorphism drawn element by element. The fibres above each point of the target all have the same size, and the fibre above the identity is the kernel.

Proposition 5.6. Let $f : G \rightarrow H$ be a homomorphism. Then $f(1) = 1$ and $f(g^{-1}) = f(g)^{-1}$ for every g .

Proof. From $f(1) = f(1 \cdot 1) = f(1)f(1)$, multiply both sides by $f(1)^{-1}$ to get $1 = f(1)$. Then $1 = f(1) = f(gg^{-1}) = f(g)f(g^{-1})$, so $f(g^{-1})$ is the inverse of $f(g)$. $\square$

Watch out

A homomorphism need not be injective or onto, and most are neither. Parity is onto but very far from injective; doubling is injective but not onto. The two special cases have names, but they are special.

Big picture

Generating sets and homomorphisms interact in the way that makes free groups important. If S generates G , a homomorphism out of G is completely determined by what it does to S , because every element is a word in S and the homomorphism property forces the value on the word. For a *free* group there is no constraint at all on those choices: any assignment of the generators to elements of H extends to a homomorphism $F_2 \rightarrow H$. For any other group, the choices must respect the relations, which is exactly the subject of Section 7.

Try it

Is $n \mapsto n + 1$ a homomorphism $\mathbb{Z} \rightarrow \mathbb{Z}$? (Check Proposition 5.6 first.) Is $n \mapsto -n$? Find a homomorphism from D_n onto $\mathbb{Z}/2\mathbb{Z}$, and describe which elements it sends to 0.

6 Subgroups, cosets and quotients

Definition 6.1. A subset H of a group G is a **subgroup** if it contains the identity, is closed under the multiplication of G , and contains the inverse of each of its elements. It is then a group in its own right.

Subgroups have appeared throughout: the rotations inside D_n , the even integers inside $\mathbb{Z}$, the inscribed polygons inside $\mathbb{Z}/n\mathbb{Z}$, the four rotations about one face axis inside the cube group. The kernel of a homomorphism, meaning the set of elements sent to the identity, is always a subgroup.

Definition 6.2. For a subgroup $H \leq G$ and an element $g \in G$, the set $gH = \{gh : h \in H\}$ is a **left coset** of H .

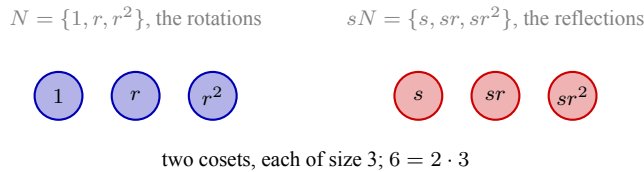


Figure 6.1 Cosets of a subgroup of order 3 inside a group of order 6: two cosets, each of size three, and together they exhaust the group. Multiplying on the left by a fixed element permutes the cosets.

Proposition 6.3. The cosets of H partition G , and all have the same size as H . Consequently, if G is finite then $|H|$ divides $|G|$.

Proof. Every g lies in gH , since $1 \in H$. If two cosets gH and $g'H$ share an element, then $gh = g'h'$ for some $h, h' \in H$, whence $g' = gh(h')^{-1} \in gH$ and therefore $g'H \subseteq gH$; by symmetry the two cosets coincide. The map $h \mapsto gh$ is a bijection from H to gH , since it is undone by multiplication by g^{-1} . So the cosets are disjoint, cover G , and all have $|H|$ elements. $\square$

It is tempting to make the set of cosets into a group by declaring $(gH)(g'H) = gg'H$. Sometimes this works and sometimes it does not, and the difference is important.

Watch out

Take $G = D_3$ and $H = \{1, s\}$, a subgroup of order 2. Using the representative s for the coset sH gives $s \cdot r = sr$, while using the representative 1 for the same coset gives a different answer, so the proposed product depends on which representative you pick. The recipe is not a well-defined rule on cosets.

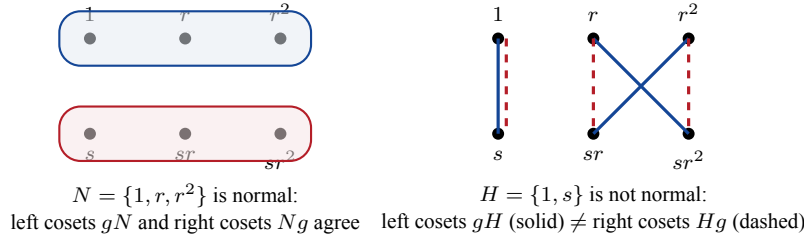


Figure 6.2 The same picture, drawn as a partition.

Definition 6.4. A subgroup $N \leq G$ is **normal** if $gNg^{-1} = N$ for every $g \in G$. When N is normal, the recipe above is well defined, the set of cosets becomes a group called the **quotient** G/N , and the map $g \mapsto gN$ is a surjective homomorphism $G \rightarrow G/N$ with kernel N .

Theorem 6.5 (First isomorphism theorem). *If $f : G \rightarrow H$ is a homomorphism with kernel N and image I , then N is normal in G and $G/N \cong I$.*

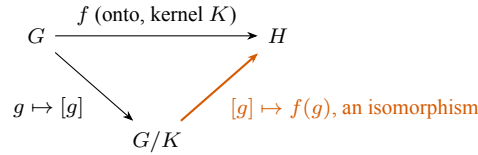


Figure 6.3 The first isomorphism theorem as a diagram: the homomorphism f factors through the quotient, and the induced map $[g] \mapsto f(g)$ is an isomorphism onto the image.

The notation $\mathbb{Z}/n\mathbb{Z}$ can now be read as what it is: the quotient of the group $\mathbb{Z}$ by the subgroup $n\mathbb{Z}$ of multiples of n . Two integers give the same coset exactly when they differ by a multiple of n , which is exactly the colouring of Section 3.3.

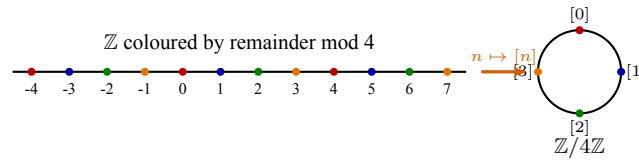


Figure 6.4 Wrapping the line round the circle: the quotient map $\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ drawn as the decoration of Section 3.4.

7 Presentations

Free groups and quotients combine into the most economical way of describing a group: name some generators, name the relations they satisfy, and stop.

Definition 7.1. Let S be a set of letters and let R be a set of words in those letters, called **relators**. The group $\langle S \mid R \rangle$ is the quotient of the free group on S by the smallest normal subgroup containing R . Concretely: elements are words in S , and two words are equal exactly when one can be turned into the other by inserting or deleting relators and by the forced cancellations $xx^{-1} = 1$.

The practical content is the “concretely” sentence. A relator is a licence: a word you are allowed to insert

anywhere, or to delete wherever it appears.

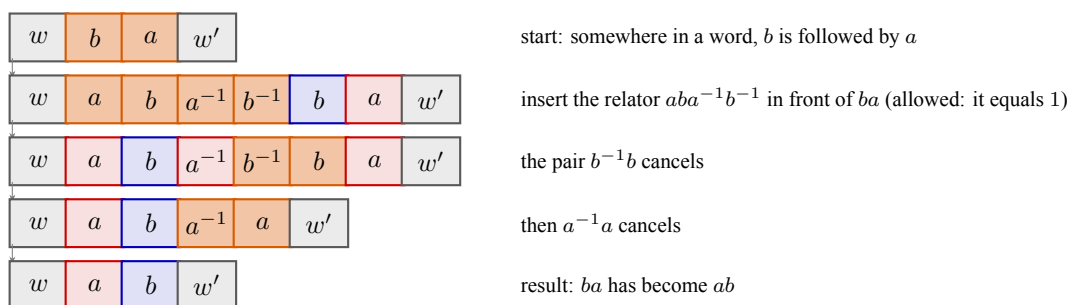


Figure 7.1 One relator, one legal move. If $aba^{-1}b^{-1}$ may be inserted and deleted, then wherever a b is followed by an a the two may be swapped.

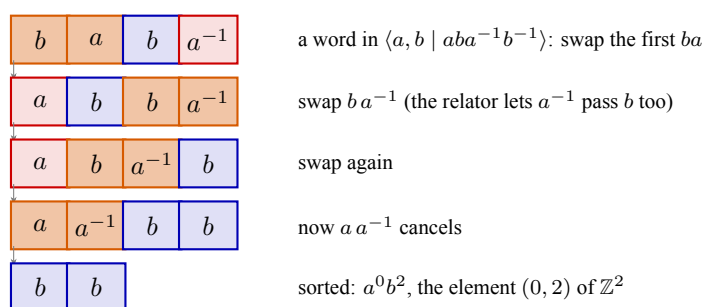


Figure 7.2 Using that move repeatedly sorts any word: push every a to the left of every b , then collect exponents by ordinary cancellation. The result is a normal form $a^m b^n$, which is exactly the element (m, n) of $\mathbb{Z}^2$.

Example 7.2. $\langle a, b \mid abab^{-1}a^{-1}b^{-1} \rangle$ is not the usual way to write it; the standard form is $\langle a, b \mid aba^{-1}b^{-1} \rangle$, and this group is $\mathbb{Z}^2$. The relator says precisely $ab = ba$, and the sorting argument above turns every word into a unique $a^m b^n$.

Example 7.3. $\langle a \mid a^n \rangle \cong \mathbb{Z}/n\mathbb{Z}$: the relator says n steps bring you home.

Example 7.4. $\langle a, b \mid \rangle \cong F_2$: no relators, no rules beyond the forced ones. This is what “free” means.

Example 7.5. $\langle s, t \mid s^2, t^2, (st)^n \rangle \cong D_n$, which is the fact promised at the end of Section 3.1. The three relators are exactly the three facts observed there.

$$\begin{array}{ccc} F(S) & \xrightarrow{\text{send each letter to itself}} & G \cong \langle S \mid R \rangle \\ \uparrow \text{kernel} & & \\ \langle\langle R \rangle\rangle = \text{normal closure of } R & & \end{array}$$

Figure 7.3 The construction behind the notation: start with the free group on S , take the smallest normal subgroup containing the relators R , and pass to the quotient.

Watch out

Presentations are treacherous. Two different-looking presentations can define isomorphic groups, and there is no algorithm that decides in general whether a given presentation defines the trivial group. Short presentations can define enormous groups. This is not a defect of the notation; it is a genuine feature of the subject, and it is one reason geometric methods are worth having.

Big picture

Every group has a presentation: take S to be any generating set, take R to be every word in S that equals the identity, and the definition above returns the group you started with. So *every group is a quotient of a free group*. Combined with the Cayley graph construction of Section 4, this is the shape of the whole subject: a group is described by generators and relations, the description is turned into a decorated graph, and questions about the group become questions about the shape of the graph.

Every presentation in this chapter, beside the group it defines and the picture that group came with. In each case a relator is a closed loop in the Cayley graph, and the shape of that graph is what the presentation is really describing.

presentation	group	its Cayley graph	drawn in
$\langle a \mid a^n \rangle$	$\mathbb{Z}/n\mathbb{Z}$	the arrowed n -gon	Figure 3.23
$\langle a \mid \rangle$	$\mathbb{Z}$	the arrowed line	Figure 3.24
$\langle a, b \mid aba^{-1}b^{-1} \rangle$	$\mathbb{Z}^2$	the grid with coloured edges	Figure 3.28
$\langle a, b \mid \rangle$	F_2	the 4-valent tree	Figure 3.39
$\langle s, t \mid s^2, t^2, (st)^n \rangle$	D_n	a $2n$ -gon with two edge colours	Problem 10

Reading a relator in the Cayley graph makes the definition concrete. A word is a path from the identity, as in Figure 3.39, and saying that a word equals the identity says that its path comes back to where it started. So the relators of a presentation are exactly the loops you are declaring to be closed, which is why the free group, with no relators, has a graph with no loops at all: a tree.

Try it

In $\langle a, b \mid aba^{-1}b^{-1} \rangle$, sort the word $ba b a^{-1}$ into the form $a^m b^n$. In $\langle a \mid a^5 \rangle$, reduce a^{13} and a^{-2} . Then explain why $\langle a, b \mid a^2, b^2 \rangle$ is infinite, even though both generators have order two.

8 Looking back: what each idea bought

This section is a consolidation. It has no new mathematics in it. Read it after Section 7, and read Figure 0.1 again alongside it.

8.1 What each idea buys

Every definition in this chapter looks like bookkeeping until you see what becomes possible with it. Here is each one, with the gap it fills, what it buys, and the smallest convincing example.

Symmetry as a rigid motion. *The gap.* “Symmetry” is a feeling until you say exactly which motions count. *What it buys.* A finite, countable, composable collection instead of a vague impression. *Example.* The cube has 24 rotations, and the number can be checked two independent ways, as 8 corners times 3 turns and as $9 + 8 + 6 + 1$ by kind of axis. Two different countings agreeing is evidence the definition is the right one.

The three axioms. *The gap.* Which collections deserve the name, and which facts hold for all of them at once. *What it buys.* Everything provable without knowing a single element: the identity is unique, inverses are unique, $gx = k$ has exactly one solution, and $gx = gy$ forces $x = y$. *Example.* The six quarter turns

of the cube look like a natural family and are not a group, because two of them in a row give a corner turn. Without closure none of the above applies.

Isomorphism. *The gap.* Two groups built from unrelated material can be the same group. *What it buys.* A dictionary, so every question about one becomes a question about the other and you may work in whichever is easier. *Example.* The rotations of the cube are S_4 . “Which turn is this?” becomes “which permutation of four things is this?”, and the second question needs no spatial imagination at all.

Cycle notation. *The gap.* A permutation written as a list of images is unreadable and hopeless to multiply. *What it buys.* A short normal form and a multiplication you can carry out with a pencil. *Example.* $(2\ 6) \cdot (1\ 2\ 4)(3\ 5) = (1\ 6\ 2\ 4)(3\ 5)$, computed by pushing six numbers through two factors.

Generating sets. *The gap.* A group can be enormous or infinite, so listing it is not an option. *What it buys.* The whole group described by a handful of elements, and a homomorphism out of it determined by where those few elements go. *Example.* Two reflections generate all ten symmetries of the pentagon, and $n - 1$ neighbour swaps generate all $n!$ permutations.

Relations and presentations. *The gap.* Words in the generators are not unique, so you must say which words are equal. *What it buys.* An entire group compressed into one line of notation. *Example.* $\langle a, b \mid aba^{-1}b^{-1} \rangle$ is $\mathbb{Z}^2$; delete that single relator and the same two generators give F_2 , a completely different group.

The free group. *The gap.* What if you impose no relations at all? *What it buys.* A universal source, since every group is a free group with relations added. *Example.* Within distance k of the identity, $\mathbb{Z}^2$ has about k^2 elements and F_2 about 3^k . Same number of generators, and the only difference is one relation.

Homomorphism, kernel, quotient. *The gap.* Comparing two groups, and building smaller ones from larger ones. *What it buys.* Structure-preserving maps, the first isomorphism theorem, and the meaning of the slash in $\mathbb{Z}/n\mathbb{Z}$. *Example.* The sign map $S_n \rightarrow \mathbb{Z}/2\mathbb{Z}$ turns a question about rearranging n objects into a question about even and odd.

The Cayley graph. *The gap.* Algebra by itself gives nothing to look at. *What it buys.* One dot per element and one coloured arrow per generator, so that words become paths and relations become closed loops. *Example.* The arrowed polygon, the arrowed line, the coloured grid and the four-valent tree are one construction applied to four groups, and all four appeared in this chapter before the name did.

8.2 Algebra making geometry computable

The chapter runs in one direction throughout. Take a question you would otherwise answer by turning an object in your hands, and convert it into a calculation you can finish on paper. Every section supplies an instance.

1. *Composing two turns of a cube.* A quarter turn about the vertical axis, then a quarter turn about the left–right axis: which single rotation is that? Each turn is a four-cycle on the long diagonals, the product of the two cycles is a three-cycle, and a three-cycle is a third of a turn about the diagonal it fixes. The algebra found the axis.
2. *Composing two reflections of a pentagon.* Reflect in one axis, then in another at 36° to it. The product is the word *st*, which is one click, so the answer is a rotation by 72° . The general rule, that reflecting in two lines meeting at θ rotates by 2θ , is read off the word.
3. *Telling the time.* Four hours after nine o'clock is $9 + 4 = 13$, then subtract 12. Everyone performs this without thinking, which is the point: $\mathbb{Z}/n\mathbb{Z}$ is the arithmetic you already do whenever a quantity wraps round.
4. *Does a walk return home?* Walk on the four-valent tree by the instructions $a b a^{-1} b^{-1} b a b^{-1} a^{-1}$. You finish where you started exactly when the reduced word is empty, so a question about an infinite picture becomes a finite cancellation.
5. *The same walk on the grid.* Use the relator $aba^{-1}b^{-1}$ to push every a past every b and collect exponents into $a^m b^n$. You are home exactly when $m = n = 0$. The same instructions can return home on the grid and not on the tree, which is the difference between $\mathbb{Z}^2$ and F_2 in one sentence.
6. *Even or odd?* Can a given rearrangement of six objects be achieved by an even number of swaps? A k -cycle costs $k - 1$ swaps, so $(1\ 6\ 2\ 4)(3\ 5)$ costs $3 + 1 = 4$ and the answer is even. A question that sounds like it needs a search is settled by one addition.

Big picture

And the axioms alone already pay. Before any example is chosen, the three axioms give: the identity is unique; inverses are unique; $(gh)^{-1} = h^{-1}g^{-1}$; $gx = k$ has exactly one solution, namely $g^{-1}k$; and $gx = gy$ forces $x = y$. Each is proved in two lines and each holds in every group in this chapter and every group you will meet later.

Looking ahead

The next office hour runs the arrow the other way. There one assumes a group acts on a space in some prescribed manner and deduces an algebraic fact about the group, with no geometry left in the conclusion. The two directions together are what the subject is made of.

9 Where this is going

The chapter began with a solid cube and ended with words and graphs, so it is worth saying plainly what has been accomplished and what comes next.

Every group in the gallery turned out to be the symmetry group of something, and in three cases the something had to be *decorated* to cut the symmetries down to size: arrows on a polygon for $\mathbb{Z}/n\mathbb{Z}$, arrows on a line for $\mathbb{Z}$, colours on a grid for $\mathbb{Z}^2$. The Cayley graph of Section 4 is the general version of that decoration, built from the group itself rather than found by luck. This is the statement the next office hour proves.

Theorem 9.1 (to come). *Every group is naturally identified with the group of symmetries of a geometric object, namely its own Cayley graph.*

That theorem is the doorway to geometric group theory. Once a group is a graph, the graph can be measured:

distances between dots, how fast the number of dots within distance k grows, whether the graph looks like a line, a plane or a tree when viewed from far away. The two examples to keep in mind are the two with the same number of generators and completely different shapes: $\mathbb{Z}^2$, whose graph is a flat grid with about k^2 dots within distance k , and F_2 , whose graph is a tree with about 3^k . A single relation, $ab = ba$, is the whole difference.

10 Problems

These follow the same order as the chapter. Hints for the starred ones are at the end.

Problem 1. Write the permutation $(1\ 3)(2\ 5\ 4)$ as a product of neighbour swaps, and check your answer by composing them one at a time. Then do the same for an arbitrary k -cycle, and deduce that a k -cycle is a product of $k - 1$ swaps.

Problem 2. The regular tetrahedron has four corners. Show that its rotation group has 12 elements, and that allowing reflections gives a group isomorphic to S_4 . Which groups arise this way from the other regular solids? Do the higher-dimensional cubes give symmetric groups?*

Problem 3. Show that m generates $\mathbb{Z}/n\mathbb{Z}$ if and only if m and n have no common factor greater than 1.* Then list all subgroups of $\mathbb{Z}/12\mathbb{Z}$ and match them to the divisors of 12.

Problem 4. Construct a group of matrices all of whose entries lie in a finite set.* How many elements does your group have?

Problem 5. Show that every word in a, b, a^{-1}, b^{-1} reduces to exactly one reduced word.* Where in the construction of F_2 is this needed?

Problem 6. Let u and v be reduced words. Show that the reduction of the concatenation uv deletes a block of letters at the seam and nothing else, and use this to compute the length of uv in terms of the lengths of u and v and the size of the cancelling block.

Problem 7. Prove the cancellation laws: in any group, $gx = gy$ implies $x = y$, and $xg = yg$ implies $x = y$. Give an example of a set with a multiplication satisfying (G1) and (G3) in which cancellation fails.

Problem 8. Show that a group in which every element satisfies $g^2 = 1$ is abelian. Then find such a group with exactly 8 elements.

Problem 9. Let $f : G \rightarrow H$ be a homomorphism. Show that the image of f is a subgroup of H and the kernel is a normal subgroup of G . Show that f is injective exactly when its kernel is $\{1\}$.

Problem 10. Draw the Cayley graph of D_3 with respect to $\{s, t\}$, its two reflections, and with respect to $\{r, s\}$, a rotation and a reflection. Compare the two pictures.

Problem 11. Show that $\langle a, b \mid aba^{-1}b^{-1} \rangle \cong \mathbb{Z}^2$ by proving that every word has a unique normal form $a^m b^n$.*

Problem 12. Show that $\langle a \mid a^n \rangle \cong \mathbb{Z}/n\mathbb{Z}$. Then decide what group $\langle a, b \mid a^2, b^3, (ab)^3 \rangle$ is, by drawing enough of its Cayley graph to recognise it.

Hints

2. Count as in Proposition 1.2: a corner can go to any of four corners, and then three rotations fix that choice. For the last part, count the rotations of the four-dimensional cube and compare with $5!$ and $6!$; the numbers do not match.
3. If $\gcd(m, n) = 1$ there are integers u, v with $um + vn = 1$; reduce modulo n . For the converse, note that every multiple of m is a multiple of any common divisor.
4. Use entries in $\mathbb{Z}/p\mathbb{Z}$ for a prime p , where every nonzero element has a multiplicative inverse, and take the invertible 2×2 matrices. For $p = 2$ there are six.
5. Induct on length, as in Theorem 3.4: two available deletions either do not overlap or overlap in one letter.
11. The relator lets you swap a b past an a . Show the swapping terminates, and that the resulting $a^m b^n$ cannot be changed further.